Clinical Pharmacology Considerations for Cell Therapy Development:

A Case Study with CARVYKTI®

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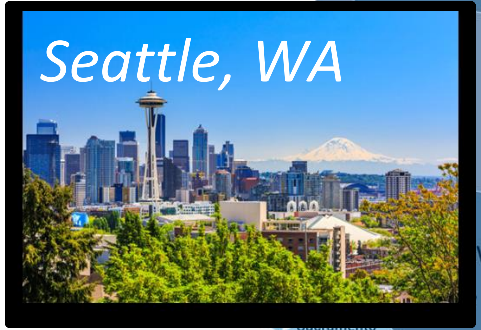
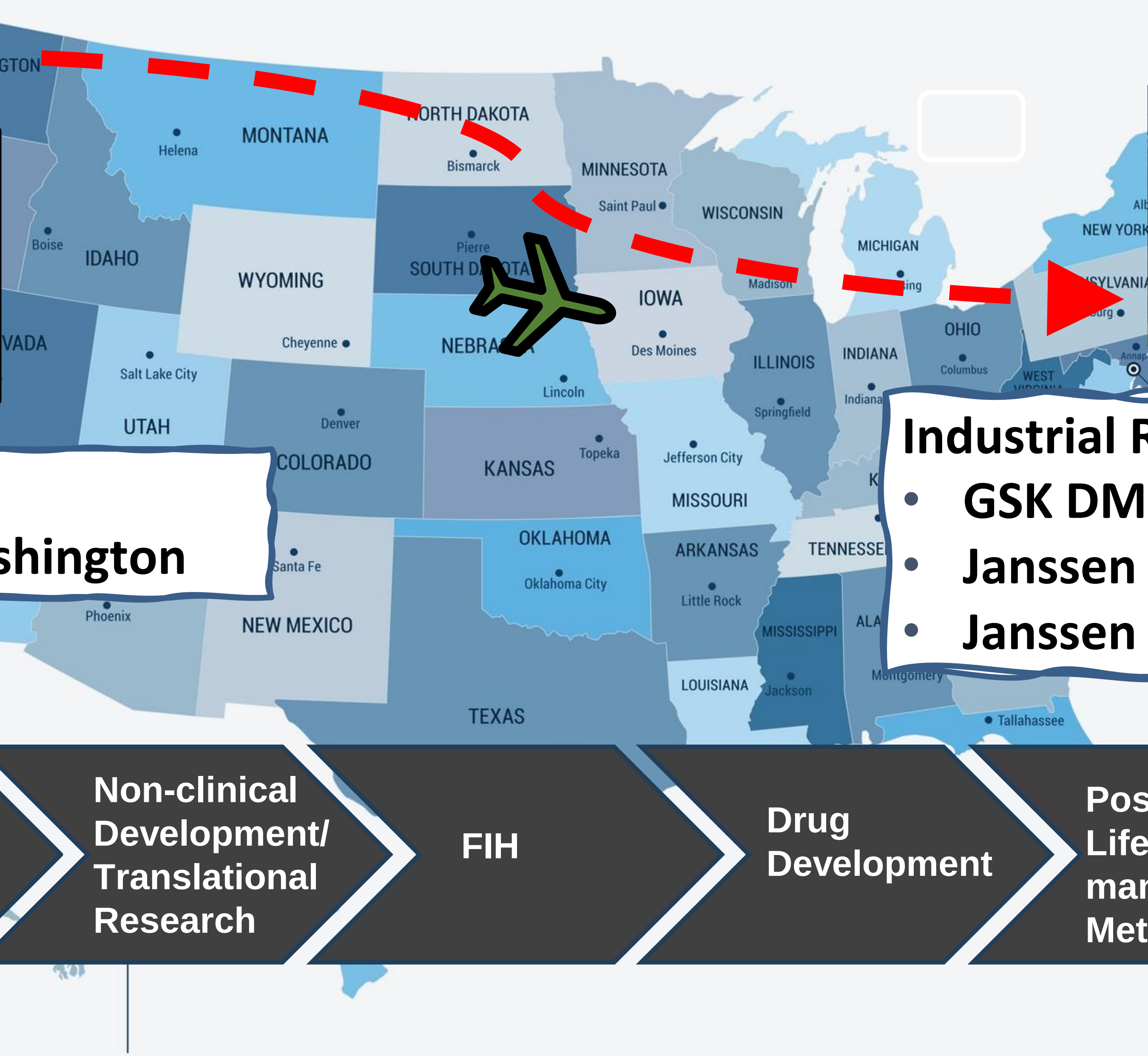
Nov 13th , 2023

Shirley Chang



PhD / Post Doc
University of Washington



Industrial R & D
- GSK DMPK
- Janssen DMPK
- Janssen ClinPharm

Drug Discovery → Non-clinical Development/ Translational Research → FIH → Drug Development → Post-approval Life-cycle management/ Meta-analysis

- **Drug Discovery**
  - Molecular Bio
  - Tox
  - ADME
- **Non-clinical Development/ Translational Research**
- **FIH**
- **Drug Development**
- **Post-approval Life-cycle management/ Meta-analysis**

Clin Pharm Related Experiences

- Academic works
  - Uremic solutes and drug Interaction:
    Quantitative assessment of chronic kidney disease (CKD) impact on renal drug clearance of tenofovir using ex vivo model and in silico model (PBPK)

Imaoka, T. et al. Sci Rep 11, 21356 (2021).
Chang., et al. Pharmaceutical Research (2023)

- Industrial works
  - FIH projects (small molecules/cell therapy within oncology TA)
    - Human dose projection/selection strategy
    - Documents review and dossier preparation for NME and IND filing
    - Delivering PK data and E-R in SET meetings

- Late development (CAR-T, CARVYKTI® sBLA submission).
  - Closeout CSR (MMY2001, NCT03548207)
  - sBLA (MMY3002, CSR, 2.7.1 & 2.7.2)

Chang., et al., ACCP poster 2023
CF de Larrea., et al., IMS 2023

Outlines

Chimeric Antigen Receptor (CAR) T-cell Therapies

- Introduction of Adoptive Cell Therapies (ACTs), including CAR T-cell therapies
- Clin Pharm Considerations for CAR T-cell therapies:
  - PK, E-R, and modeling approaches based on approved products in the clinic
  - Clinical pharmacology strategies for the future of cell therapy discovery and development
- Case study: CARVYKTI® (CARTITUDE-1 (phase1b/2), CARTITUDE-4 (Phase 3) for BLA)
- Challenges/opportunities in CAR T-cell therapy development
- Summary/Conclusion/Acknowledgment/Q & A

Lessen learned from FIH studies (small molecules)

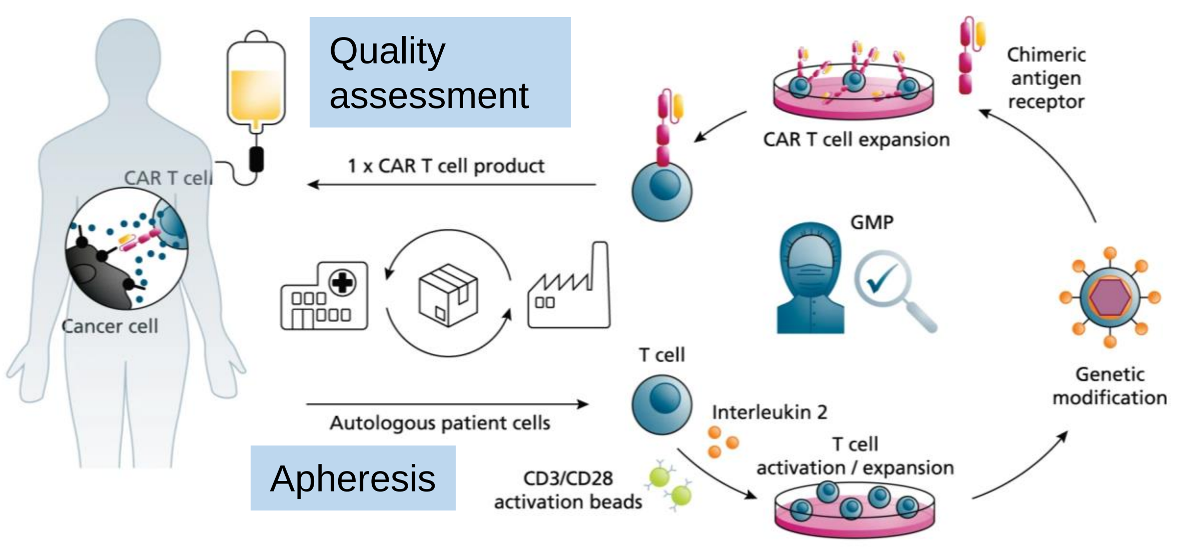
- How can Model-Based Drug Development (MBDD) help in early decision-making for FIH studies? Compound D an example

Introduction

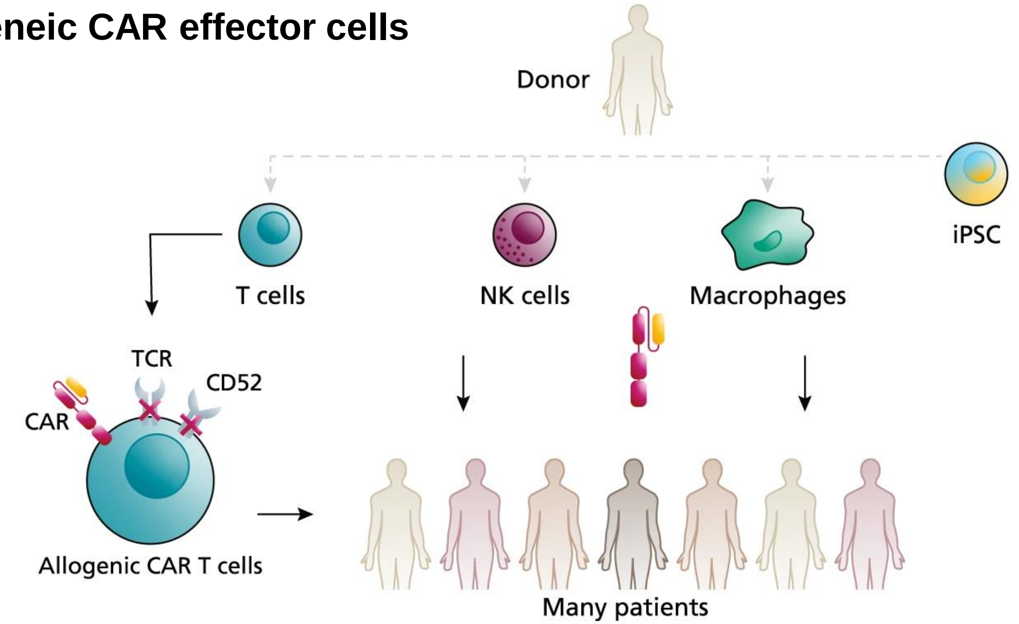
- Chimeric Antigen Receptor (CAR) T-cell Therapies
  - → Adoptive Cell Therapies (ACT)
  - → Human gene therapies[1]
- Other modified lymphocyte therapies
  - → CAR Natural Killer (NK) cells
  - → T cell receptor (TCR)-modified T-cells
- Autologous vs. allogeneic products
  - → T cells from the same individual patient
  - → Cells from a healthy donor

[1] Considerations for the Development of Chimeric Antigen Receptor (CAR) T Cell Products. Guidance for Industry. 2022

Autologous CAR T cells



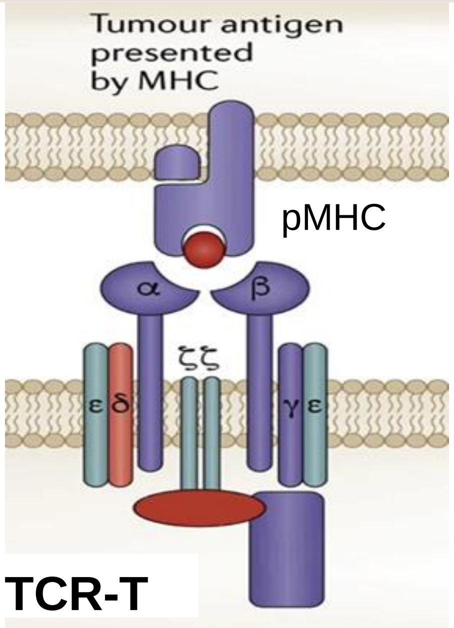
Allogeneic CAR effector cells

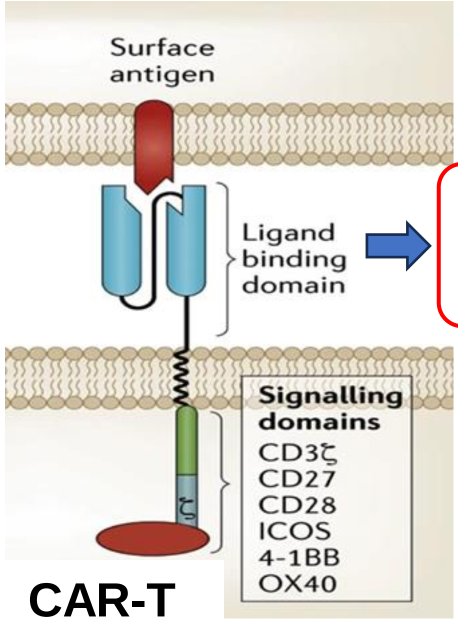


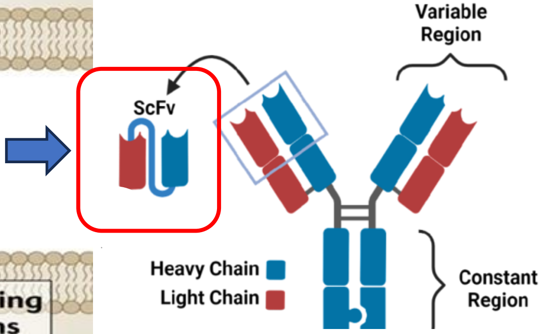
Blache, U. et al., Nat Commun 13, 5225 (2022).

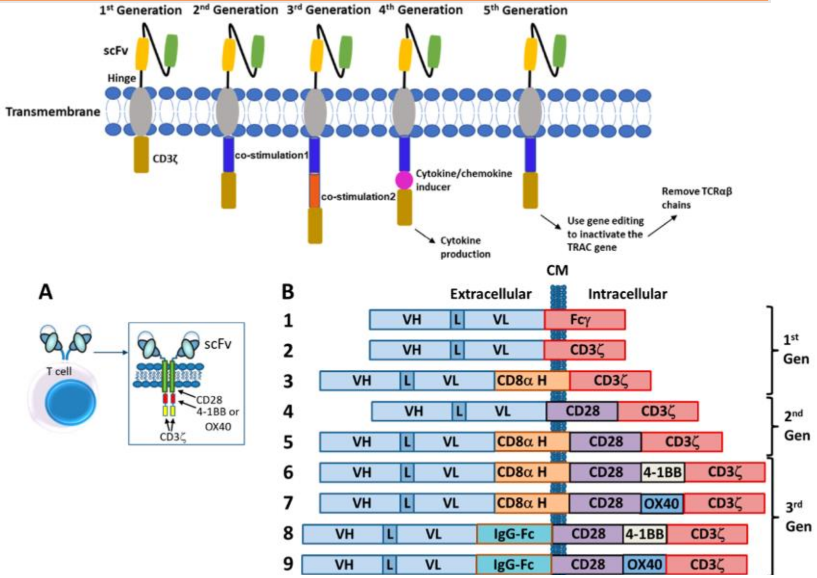
TCR-T vs. CAR-T

Garber, K.
Nat Biotechnol 36, 215–219 (2018









CAR structure components:

TCR-T

- Targets <u>intracellular</u> proteins presented as peptide-MHC (pMHC)[1]
- MHC dependent and HLA restricted
- Lower affinity, higher avidity, lower antigen density requirement[2,3,4].
- Advantage in solid tumors.

MHC: Major Histocompatibility Complex
HLA: Human Leukocyte Antigen

CAR-T

- ScFv antibody-like binding
- Largely limited to <u>cell surface antigens</u>
- MHC-independent and no HLA restriction.
- Requirement higher affinity/antigen density[2,3].
- Well suited for heme.

scFv: single chain fragment variant

CAR structure components

- Antigen recognition (extracellular) domains: scFv
- Hinge (transmembrane) domains
- Signaling intracellular domains (immunoreceptor tyrosine-based activation motifs, ITAM): CD3ζ
- Co-stimulatory domains: e.g., CD28, 4-1BB (CD137), and CD40

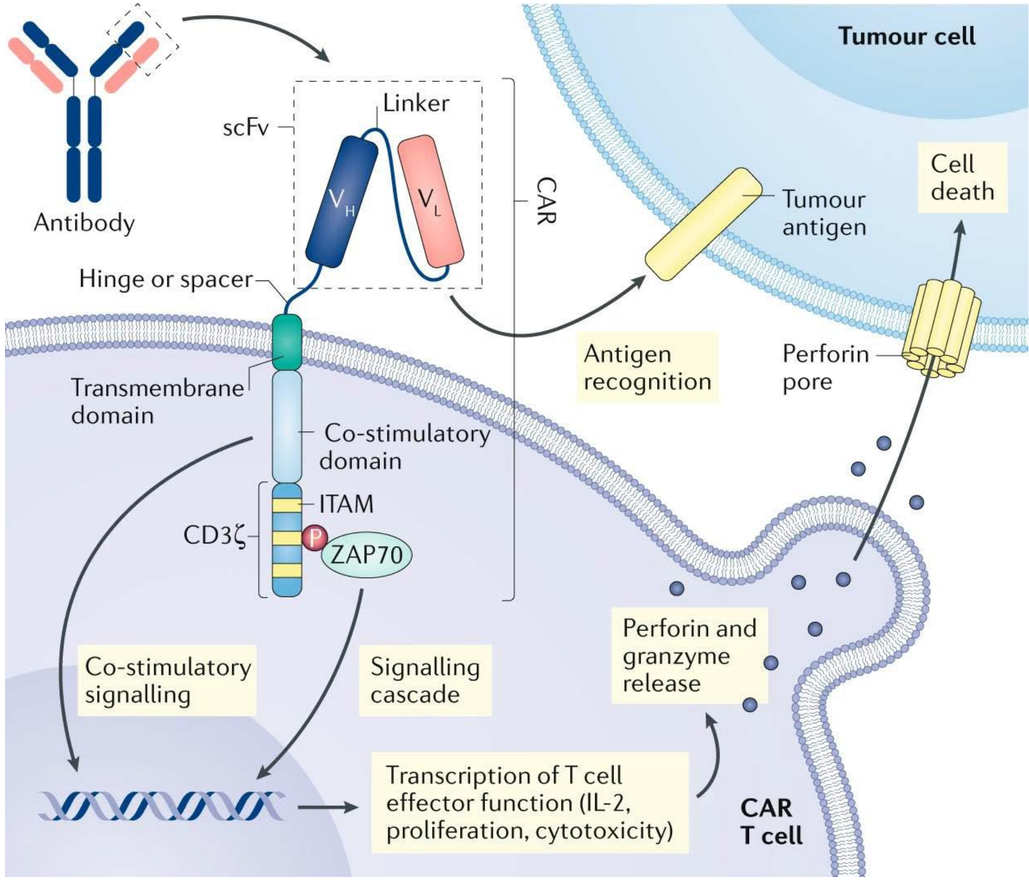
5th gen or next: Using a gene editing approach to reduce the risk of graft-versus-host disease ((GvHD) or including suicidal genes to trigger depletion in case of toxicity[6]

[1]Dubrovsky *Oncoimmunology* 2016.
[2]. *Strohl et al., Antibodies, 2019*
[3]*Harris, J Immunol, 2018*
[4]*Kalergis, Nat Immunol , 2001.*
.
[5] *Kast, J et al. Clin Transl Sci. 2022*
[6] *Tokarew, N et al. Br J Cancer. 2019*

Background – Mechanism of Action (MoA)

- Antigen recognition and engagement
- Induction of signaling cascade in T-cells
- T-cell activation and proliferation
- Perforin and granzyme release from activated T-cell; and cytokine production
- Tumor cell killing through perforin and granzyme activity

Mechanism of action for a 2nd-generation CAR-T therapy[1]



[1] Larson, RC et al. Nat Rev Cancer. 2021

US Approved Products

- Six CAR-T oncology products (autologous) have been approved in the US
- Accelerated approval for cell and gene therapies is incoming

Brand name (generic name)	Approved indications	Target	Year of first approval in the US	Approved dose (US)
CARVYKTI (ciltacabtagene autoleucel)	Adult R/R multiple myeloma	BCMA	2022	0.5–1 $\times$ 10^6/kg, maximal of 100 $\times$ 10^6 cells
ABECMA (idecabtagene vicleucel)	Adult R/R multiple myeloma	BCMA	2021	300–460 $\times$ 10^6 cells
BREYANZI (lisocabtagene maraleucel)	Adult R/R large B-cell lymphoma	CD19	2021	3L+: 50–110 $\times$ 10^6 cells; 2L: 90–110 $\times$ 10_6 cells
TECARTUS (brexucabtagene autoleucel)	Adult R/R mantle cell lymphoma; adult R/R B-cell precursor acute lymphoblastic leukemia	CD19	2020	r/r MCL: 2 $\times$ 10^6/kg, maximum of 200 $\times$ 10^6 cells; r/r B-cell ALL:1 $\times$ 10^6/kg, maximum of 100 $\times$ 10^6 cells
YESCARTA (axicabtagene ciloleucel)	Adult R/R large B-cell lymphoma; adult R/R follicular lymphoma	CD19	2017	2 $\times$ 10^6/kg, maximum of 200 $\times$ 10^6 cells
KYMRIAH (tisagenlecleucel)	Adult R/R large B-cell lymphoma and follicular lymphoma; pediatric refractory acute lymphoblastic leukemia	CD19	2017	Pediatric and young adult: 0.2–5.0 $\times$ 10^6/kg ($\leq$ 50 kg); 10–250 $\times$ 10^6 cells (> 50 kg); Adult: 60–600 $\times$ 10^6 cells

Abbreviations: BCMA, B-cell maturation antigen; CD19, cluster of differentiation 19; R/R, relapsed or refractory.

Translational Research and Clinical Pharmacology Considerations in ACTs

Product/Patient Factors → Cellular Kinetics (PK) /Distribution/Biomarkers → **Key Drivers Of Clinical Outcome**

- **Response (ORR,CR)**
- **Safety (CRS, NE)**

NE, neurological event

- Correlations between cellular kinetics and outcomes

Potential key driver of safety/efficacy	Associated control point(s)
Dose and regimen	• Number of cells infused (engineered and total) eg: target dose range: 0.5–1 × 10^6/kg • PK characterization • Exposure-response analysis
Composition of the infusion product (eg, cell phenotype, polyfunctionality)	T-cell engineering (vector copy number), ex vivo manufacturing process for cell selection, stimulation, and expansion. • Out-of-specification (OOS) or comparability studies (manufacturing process changes)
Patient/tumor characteristics that favor benefit-risk	Patient selection based on target expression, tumor burden, demographics, and other tumor characteristics/biomarkers. • Effect of intrinsic factor (PopPK and other modeling approach) • PD biomarker
In vivo homeostasis and other signals favoring expansion and persistence	T-cell engineering, optimization of lymphodepletion. • Immunogenicity (anti-drug antibody (ADA))

Mody, H et al., Clin Pharmacol Ther 2023, 114: 530-557

Characteristics of CAR T-Cell Therapy (Autologous Platform)

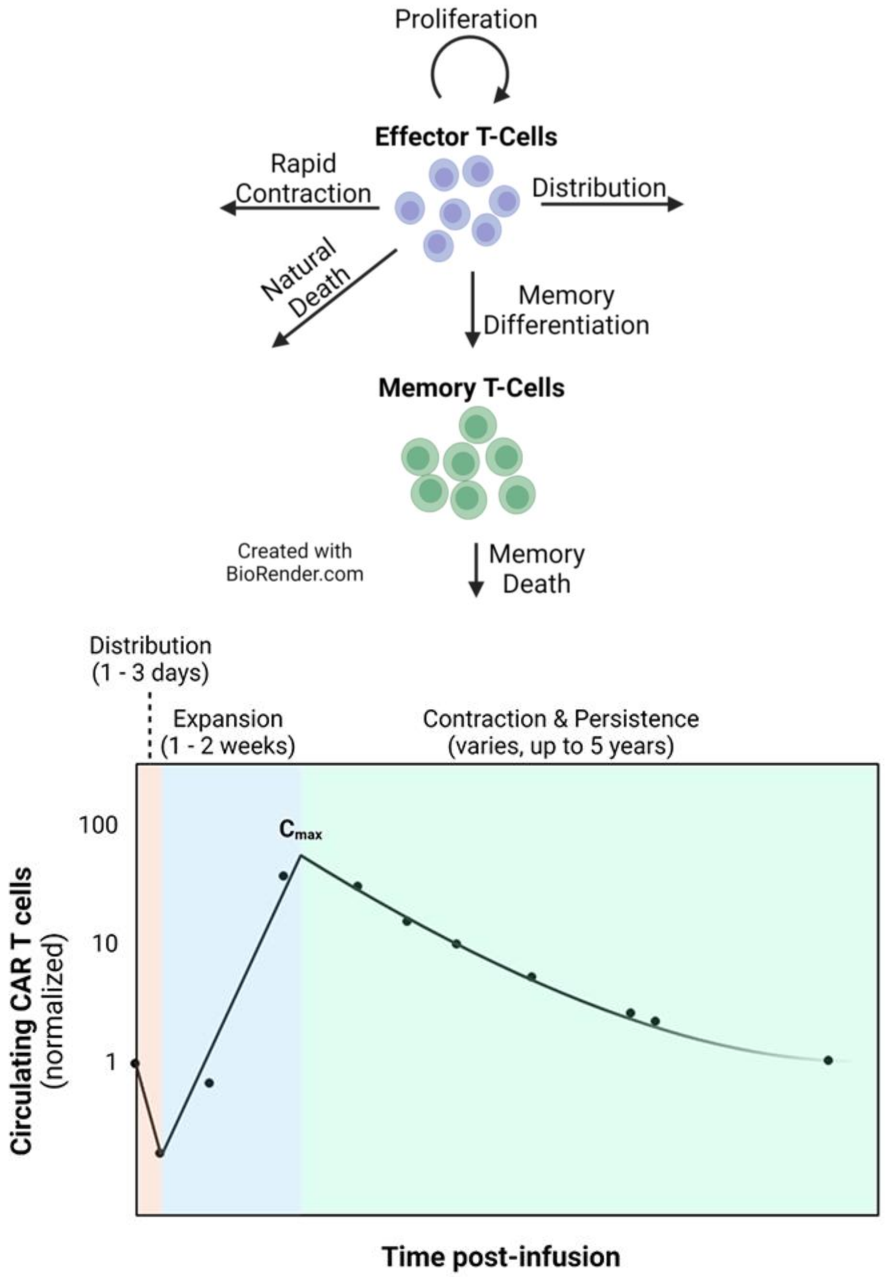
- PK
- Dose-exposure relationship
- Exposure–response analysis for efficacy and safety
  - CAR-T AE mitigation strategies
- Modeling Approaches
- Immunogenicity (ADA) - case study

Pharmacokinetics (PK) of CAR-T Therapies

- Defined as "Cellular Kinetics (CK)"

  → Different mechanism but similar C-t curve (multiphasic)

  - **Margination/distribution**: T-cell clones/phenotypes in circulation
  - **Expansion:** in response to target and homeostatic cytokines
    - Maybe driven by clonotypes
    - Unlike TMDD for mABs, target engagement generally increases circulating cell levels
  - **Contraction/Redistribution**
    - Some effector T-cells become exhausted and die
  - **Persistence**
    - Other T- cells differentiate to memory T cells and persist
    - May provide longer-term disease control



Qi et al, Adv Drug Deliv Rev (2022) 188:114421

TMDD: Target mediated drug disposition

Pharmacokinetics (PK) of CAR-T Therapies

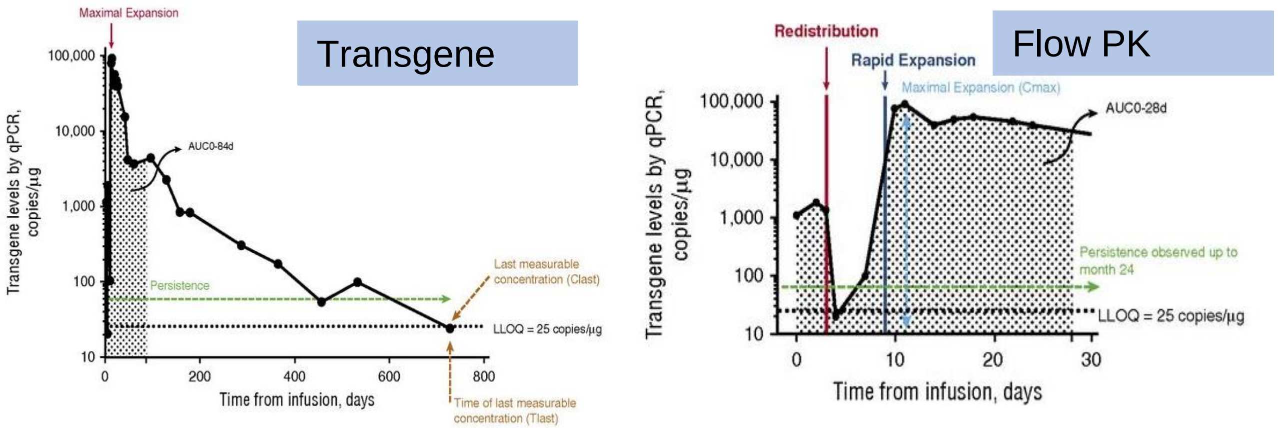
- Measurements ("concentrations") of PK or CK of CAR-T therapies
  → CAR transgene level in cells measured by qPCR: Units - CAR DNA copies/μg of genomic DNA
  → CAR surface expression on cells measured by flow cytometry: Units - cells/μl or %

- Analysis of dose-exposure relationship
  → NCA analysis is applicable and usually used
  → Typical compartmental analysis is not relevant
  → Typical PK concepts (e.g., CL, V) are not applicable

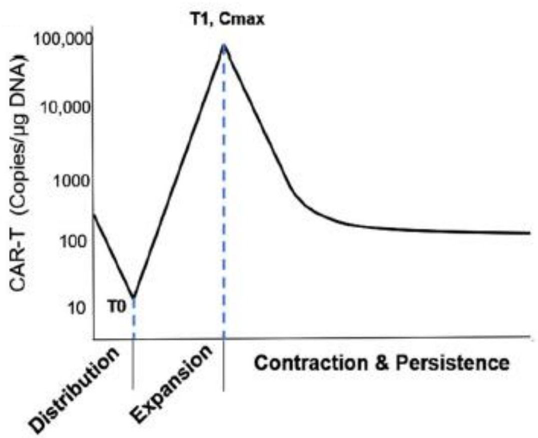
- Exposure metrics
  → For expansion phase: C_{max}, T_{max}, and $AUC_{0\text{-}28days}$
  → For persistence phase: C_{last}, T_{last} (T_{bql}), terminal $t_{1/2}$ and AUC_{last}

PK Profiles of CAR-T therapy

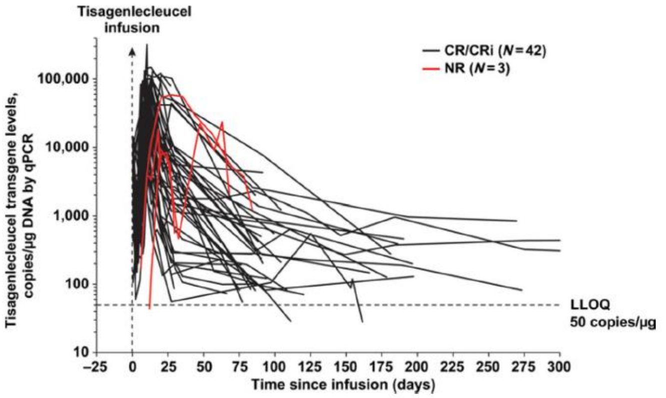
A: a representative profile from a patient with B-ALL after infusion of KYMRIAH[1]



B: a typical multi-phase kinetics[2]



C: individual level profiles of KYMRIAH[3]

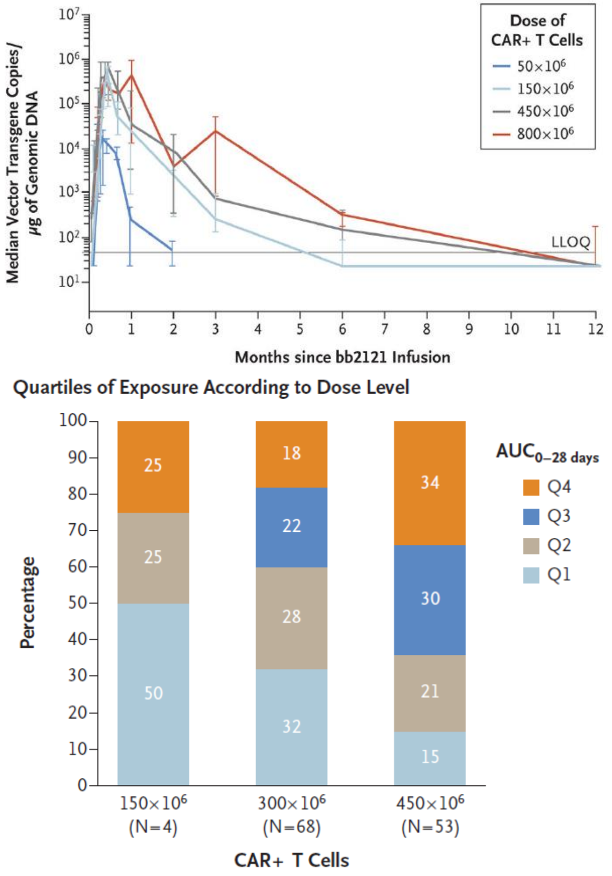


[1] Mueller, KT et al. Blood. 2017
[2] Liu, C et al. Clin Pharmacol Ther. 2021
[3] Mueller, KT et al. Clin Cancer Res. 2018

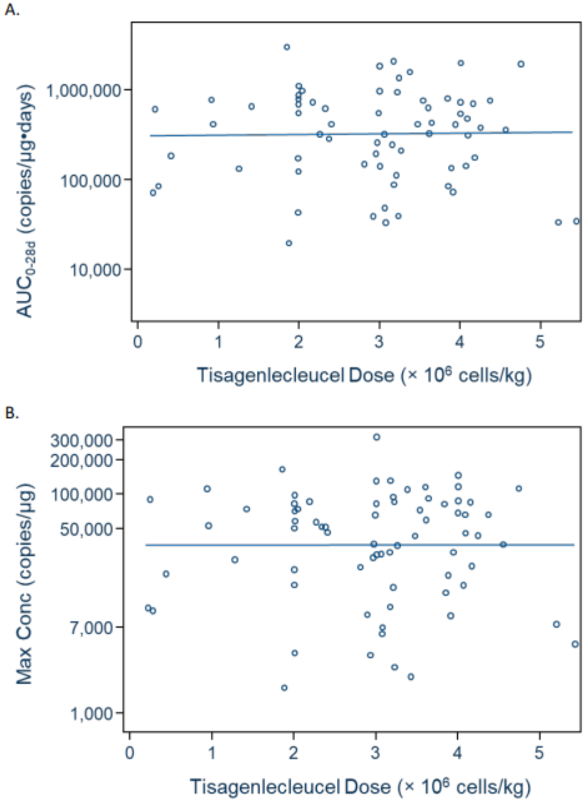
Pharmacokinetics (PK) of CAR-T Therapies

- Lack of consistent Dose-Exposure relationships: high variability and overlapping exposures across wide dose levels (CAR-T cell dose poorly correlates with cellular kinetics parameters)

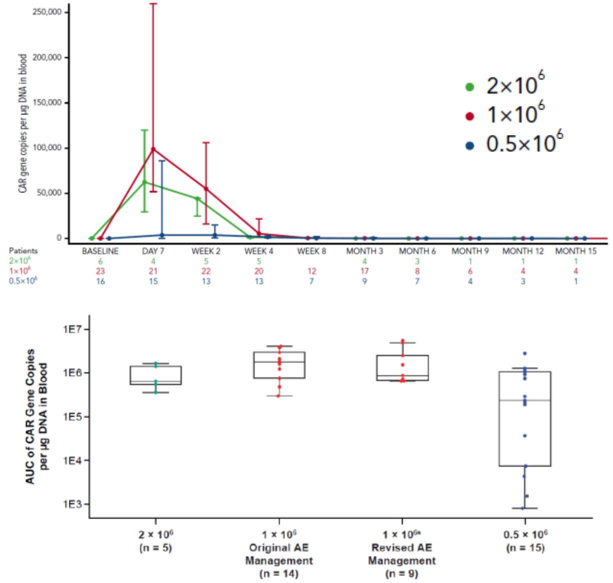
ABECMA in multiple myeloma[1,2]



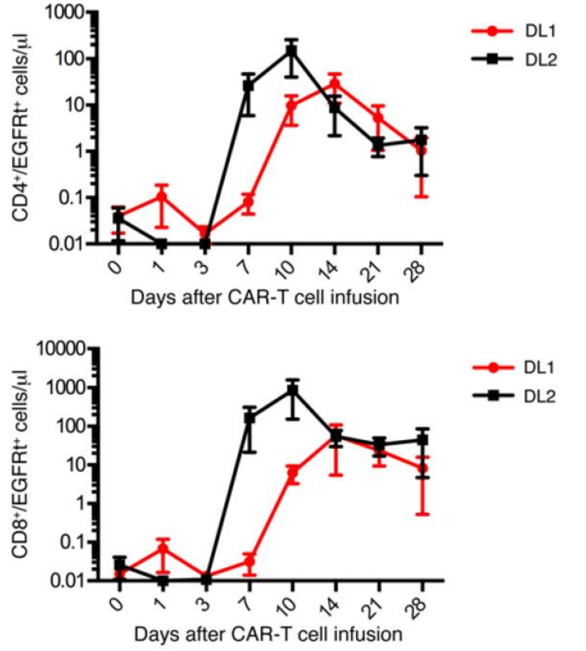
KYMRIAH in lymphoblastic leukemia[3]



TECARTUS in lymphoblastic leukemia[4]



BREYANZI in lymphoblastic leukemia[5]



DL1: 2 x 10^5 cells/kg
DL2: 2 x 10^6 cells/kg

[1] Raje, N et al. N Engl J Med. 2019
[2] Mushi, NC et al. N Engl J Med. 2021
[3] Maude, SL et al. N Engl J Med. 2018
[4] Shah, BD et al. Blood. 2021
[5] Turtle, CJ et al. J Clin Invest. 2016

Dose-Response Relationship of CAR-T Therapies

Efficacy

→ Outcome: ORR, DOR, PFS, OS, and minimal residual disease (MRD, for multiple myeloma)

→ Generally, consistent exposure-efficacy relationship seen responders vs. non-responders

I. Higher exposure (higher C_{max}, $AUC_{0\text{-}28d}$)/short expansion time (lower T_{max})/longer persistence time (higher $t_{1/2}$ and T_{last})

II. PFS, as well as depth (MRD negative) and durability of responses also shown to be associated with exposure

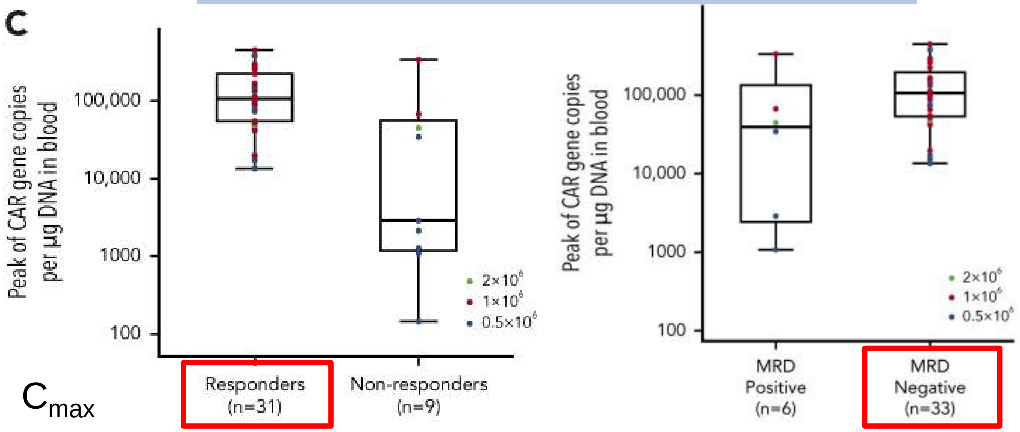
Safety

→ TEAE: cytokine release syndrome (CRS), neurologic events (NE) and hematologic toxic effects (e.g., neutropenia, anemia, thrombocytopenia, leukopenia, and lymphopenia)

→ A trend that patients with more serious AE (CRS or NE) have higher exposure (median C_{max} or AUC)

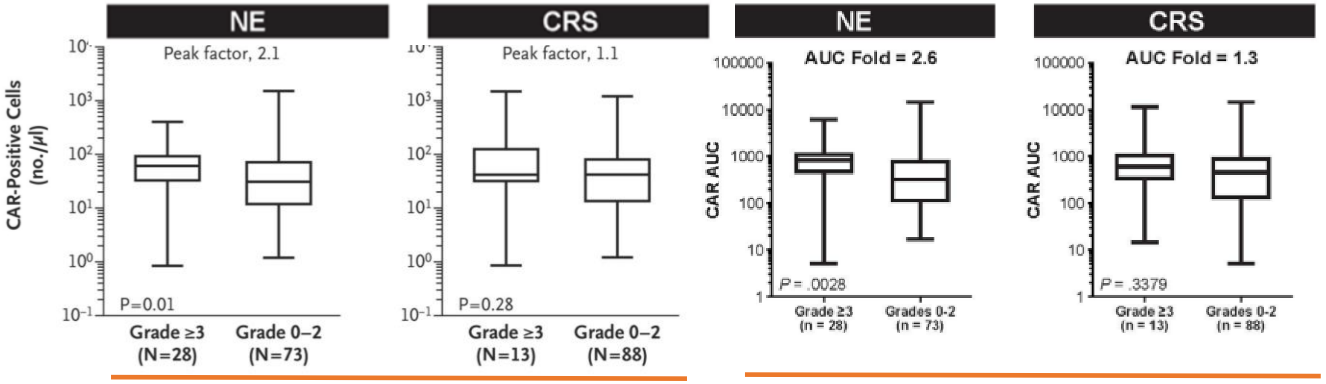
ICANS: immune effector cell-associated neurotoxicity
MNT: movement and neurocognitive TEAE
(Parkinsonism and Guillain-Barré syndrome)

TECARTUS in lymphoblastic leukemia[1]



C_{max}

YESCARTA in lymphoma[2]



C_{max} $AUC_{0\text{-}28d}$

[1] Shah, BD et al. Blood. 2021

[2] Neelapu, SS et al. N Engl J Med. 2017

CAR-T AE Mitigation Strategies

The median time to onset of CRS was 7 days (range: 1 to 12 days)
The median time to onset of ICANS was 8 days (range: 1 to 28 days)

→ PK expansion (C_{max} and $AUC_{0\text{-}28d}$)

FDA-mandated training as part of a Risk Evaluation Mitigation Strategy (REMS)

- Patient Information and Education
- Inpatient or close monitoring (at least 4 weeks after CAR-T infusion)
- Treatment: tocilizumab (anti-IL-6) and corticosteroids

MB Marzal-Alfaro et al., Front. Oncol., 11 March 2021

Modeling Approaches for CAR-T

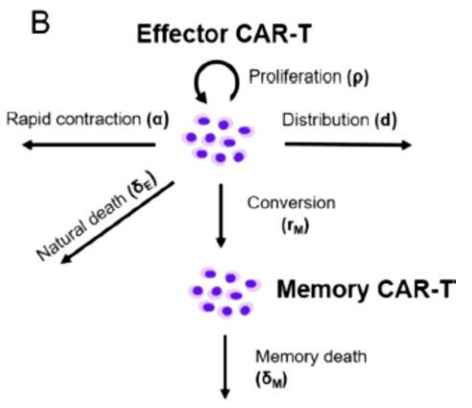
More empirical — **Fit-for-purpose** — More mechanistic

Empirical
E vs. M
No tissue

Semi-mech
E vs. M and cytokine included
Lumped tissue

QSP/PBPK:
More cell subsets (T_N)
Tissue of interest
Biomarker (target related: BCMA)

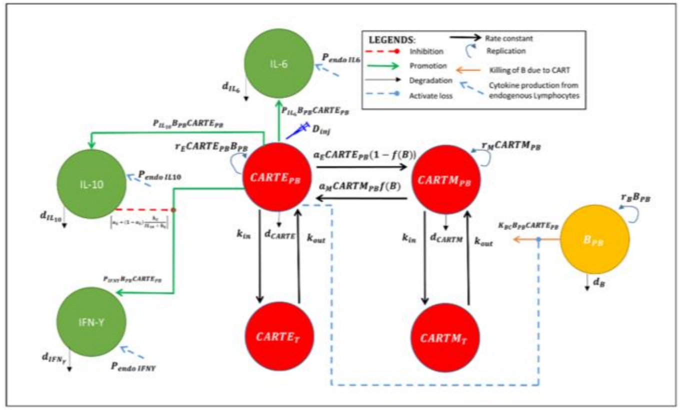
Modified model[1]



$$\frac{dE}{dt} = -\alpha \cdot E - \delta_E \cdot E - r_M \cdot E, when\ T_1 < t$$

$$\frac{dM}{dt} = r_M \cdot E - \delta_M \cdot M, when\ T_1 < t$$

[1] Liu, C et al. Clin Pharmacol Ther. 2021



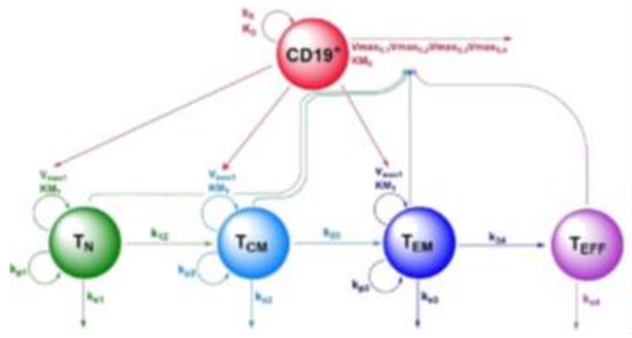
The model(s) could

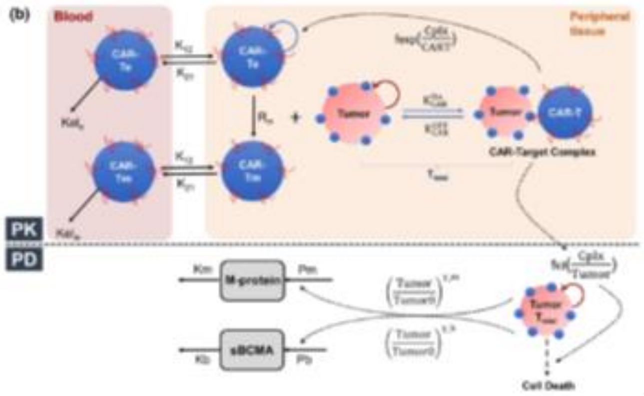
→ Capture clinical multi-phasic PK (CK) profile

→ Potentially test the impact of extrinsic and intrinsic factors and the impact of CRS-treating therapies

→ Explore dose-exposure-response relationships

[2] Hardiansyah, D et al. Clin Transl Sci. 2019





[3] Mueller-Schoell, A et al. Cancers. 2021
[4] Singh, AP. CPT Pharmacometrics Syst Pharmacol. 2021

Why Mechanistic Models?

- Integrates rich data.
- Test hypotheses in silico to reduce clinical studies.
- Reverse translation.
- Prioritize candidates in early development.
- Inform dose/schedule for combinations.

Summary of CAR T-cell (Autologous Platform) Characteristics

- Different mechanisms and profile of PK (CK) from small or large-molecule drugs (in vivo cell proliferation and differentiation)
- Lack of consistent Dose-Exposure (Dose-Response) relationships partially due to high variability
- Clinical response (efficacy and safety) generally associated with exposure (expansion: C_{max} and $AUC_{0\text{-}28d}$)
- Potential future directions of clinical pharmacology and pharmacometrics for CAR-T therapies

  → Mechanism-based modeling of PK (CK) and PD of CAR-T therapies (e.g., QSP model)

  → Deeply quantitative dose-exposure-response characterization and factors impacting

  → Further preclinical to clinical translation

Translational and Clinical Pharmacology Considerations/Strategies in Cell Therapy Drug Development (CAR-T autologous platform)

PRECLINICAL		CLINICAL			
Candidate Selection	Preclinical Development	Dose Finding Phase 1a	Proof of Concept Phase 1b/II	Confirmatory Phase III	Approval/ Post-Marketing

Preclinical:

- Empirical approach: allometrically scale CAR-T doses from preclinical models to humans
- Benchmark CAR-T approach: determined from prior preclinical and clinical data
- Translational approach to evaluation cell proliferation/killing potency/ cytokine induction in vitro: In vitro: effector to target (E:T) ratios
- Bioanalytical method development

Clinical:

- PK characterization: expansion, contraction, and persistence
- Dose and schedule selection[1] (C_{max} and $AUC_{0\text{-}28D}$): Safety management to cover DLT and onset of TEAE (CRS and NEs)
- Effect of intrinsic and extrinsic factors: BW, race and age and effect of extrinsic factors; bridging therapy (type and cycle number) and comparability of altered manufacturing process (Pop PK)
- Immunogenicity and its impact on safety and efficacy: no impact so no need to develop NAb assay
- Bridging therapy, lymphodepletion regimen and maintenance optimization
- Combination strategy (mechanistic model: QSP applied to T-cell subset)

Approval/Post-Marketing:

- Decision-making (manufacturing process)
- Drug label
- Long-term follow-up for gene therapy [2]

1 Consideration for the development of chimeric antigen receptor (CAR) T cell products–draft guidance for industry. (2022). Accessed April 17, 2023.
2 LTFU after administration of human gene therapy products–guidance for industry. (2020). Accessed April 17, 2023.

FIH Study Design Considerations for Cell Therapy

Caveats:

- lack of relevant animal models to assess CRS or NE, which is commonly observed in the clinic
- Dose- exposure is not clear in the clinic

Pros:

- Higher doses may not necessarily link to better efficacy (traditional oncology FIH study dose escalation designs for finding MTD doesn't applied)
- Target-related toxicity, such as CRS and NE is overall well-managed based on clinic experiences (close monitoring after infusion for at least 4 weeks)

FIH trial design

- Establishing a toxicity and efficacy probability interval (TEPI)
- Bayesian Optimal Interval (BOIN)-guided escalation strategy (clear dose escalation and de-escalation strategy)

Legend-2 study (China)

1 Consideration for the development of chimeric antigen receptor (CAR) T cell products–draft guidance for industry. (2022). Accessed April 17, 2023.
2. IQ consortium white paper: Mody, H et al., Clin Pharmacol Ther 2023, 114: 530-557

Case Study: CARVYKTI®
(ciltacabtagene autoleucel;cilta-cel)

CARTITUDE-1 (MMY2001: phase1b/2) and CARTITUDE-4 (MMY3002: Phase3) for BLA

- Trial design
- Cilta-cel PK profiles (transgene and CART+ cell count)
- Exposure-response analysis (efficacy/safety)
- Popk model (empirical model)
- Immunogenicity: anti-drug antibody (ADA)
- PD biomarker: soluble BCMA (sBCMA)

LEGEND BIOTECH | Janssen Oncology

CARTITUDE

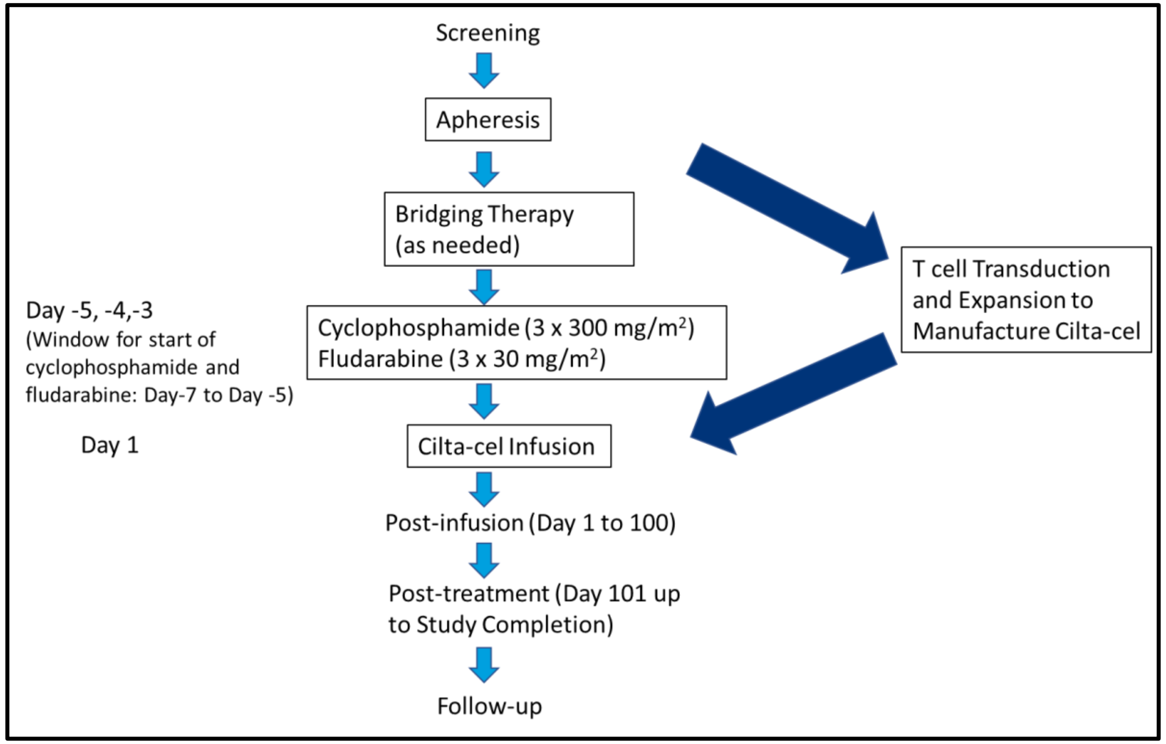
CARTITUDE Studies: CART-1 and CART-4

CARVYKTI®: CAR-T therapy directed against B-cell maturation antigen (BCMA)

Indication: Relapsed or Refractory Multiple Myeloma

Dose: single dose
Total targeted dose of 0.75 x 10^6 CAR-positive viable T cells/kg (range: 0.5-1.0 x 10^6 CAR-positive viable T cells/kg)

Schematic Overview of GRATITUDE -1/CARTITUDE-4



<u>CART-1 (BLA) Proof of Concept:</u>
Phase1b-2, Open-Label Study (BLA)

Populations: <u>R/R MM at least 3 PL of **PI and IMiD</u>, N=97

- ORR = 93.8% (87.0%, 97.7%)
- CR or better (sCR + CR) = 76% (68.8%, 86.1%)
- The 30-month OS survival rate was 68.0% (95% CI: 57.7-76.3)

<u>CART-4 (sBLA) Confirmatory:</u>
Phase 3 Randomized Study (sBLA):

- Arm A: *PVd or DPd, N= 208
- Arm B: Cilta-cel, N=208 (N=176 for study treatment)

Populations: Relapsed and lenalidomide-refractory MM
<u>Received 1-3 PL of **PI and IMiD</u>

- ORR = 99.4% (96.9%, 100.0%)
- CR or better (sCR + CR) = 86.4% (80.4%, 91.1%)

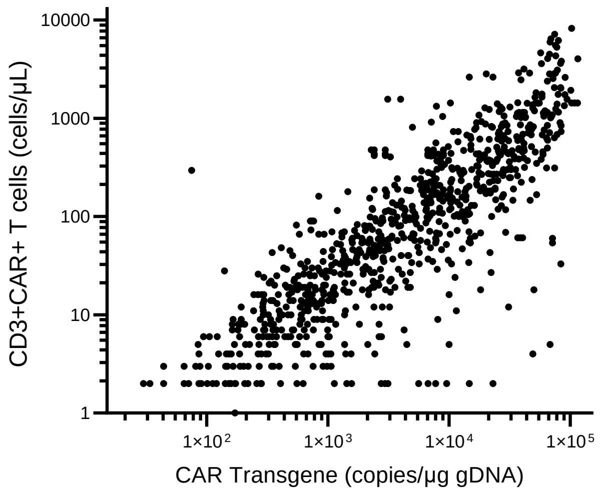
*Pomalidomide, Bortezomib and Dexamethasone (PVd) or Daratumumab, Pomalidomide and Dexamethasone (DPd) ,
**proteasome inhibitor (PI)
immunomodulatory drug (IMiD)

PK Measures are Concordant in CART-1 (MMY2001) and CART-4 (MMY3002)

Transgene by qPCR and CD3+CAR+ T cell count by flowcytometry

CART-1 (MMY2001)

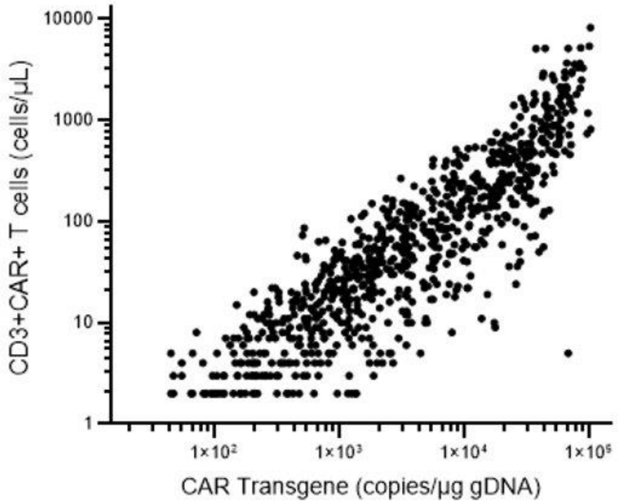
- Transgene blood samples were measured by qPCR (LLOQ = 50 copies/µg gDNA)
- CD3+CAR+ T cells blood samples were measured by flow (LLOQ= 2 cells/µL)



Pearson r =0.402
P<0.0001
N=2203

CART-4 (MMY3002)

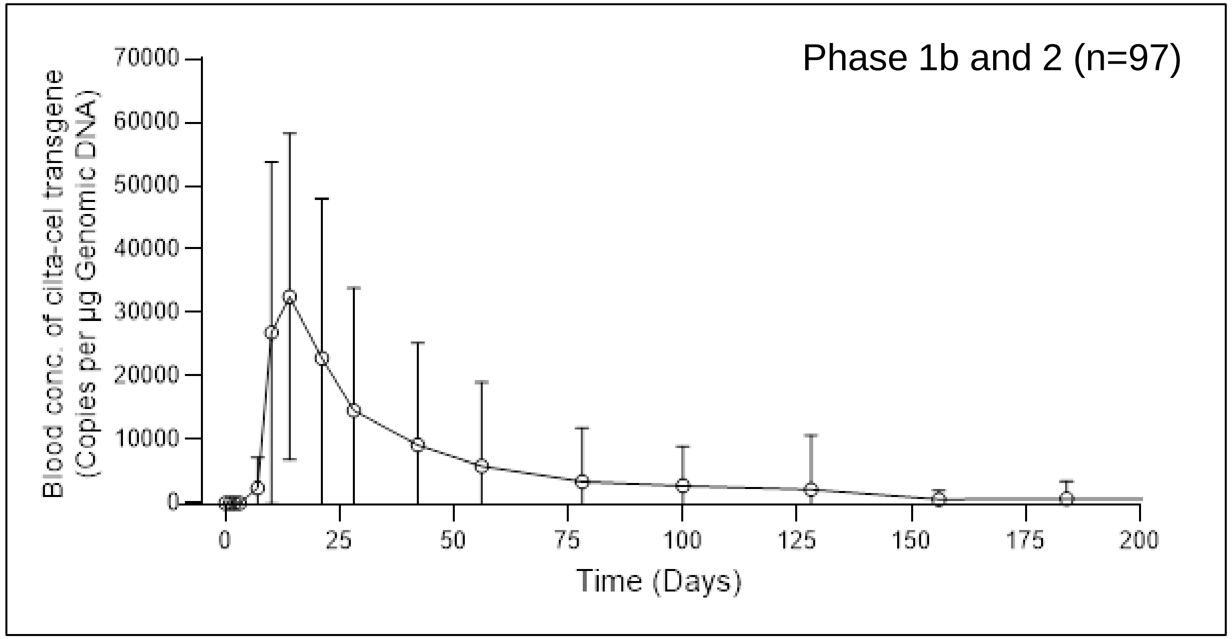
- Transgene blood samples were measured by qPCR (LLOQ = 25 copies/µg gDNA)
- CD3+CAR+ T cells blood samples were measured by flow (LLOQ= 2 cells/µL)

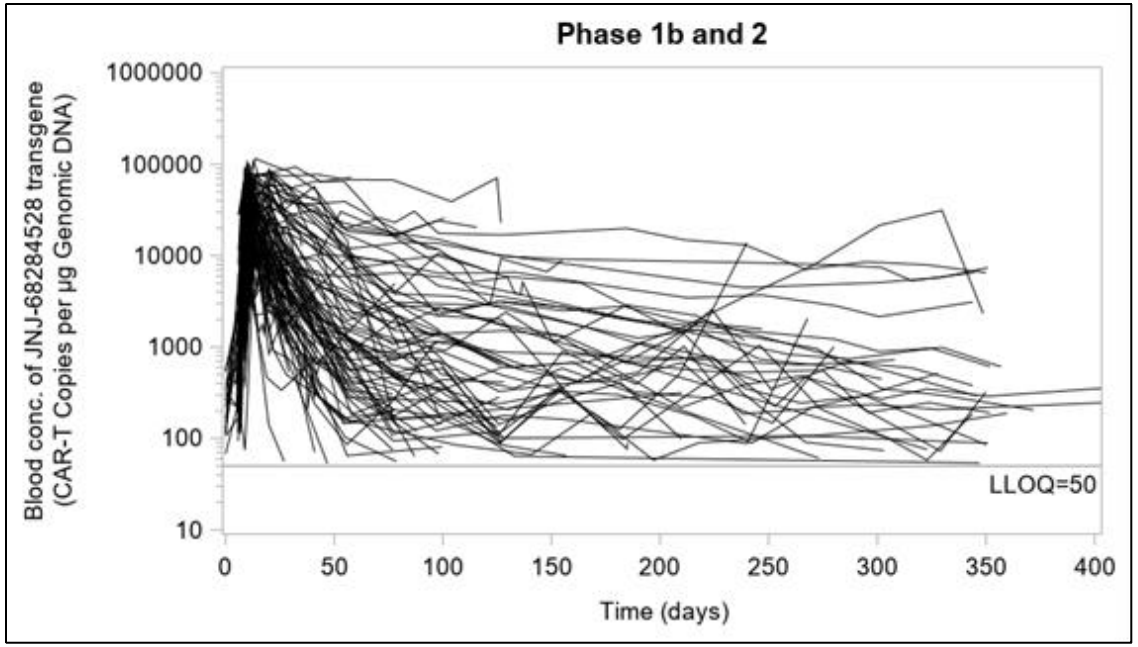


Log transformed
Pearson r= 0.879
P<0.0001
N=826

CAR Transgene Concentration Profile in CART-1

- Blood transgene conc showed high intersubjective variability (%CV= 125.2% for AUC)





1. Expansion
2. Contraction
3. Persistence

Safety management (CRS and NE)

The median time to onset of CRS was 7 days (range: 1 to 12 days).
The median time to onset of ICANS was 8 days (range: 1 to 28 days).
NE: The median onset of parkinsonism in the 5 patients in CARTITUDE-1 was 43 days (range: 15 to 108 days)/Median time to onset of cranial nerve palsies was 26 days (range: 21 to 101 days)

PK Metrics Are Comparable Between CART-1 and CART-4

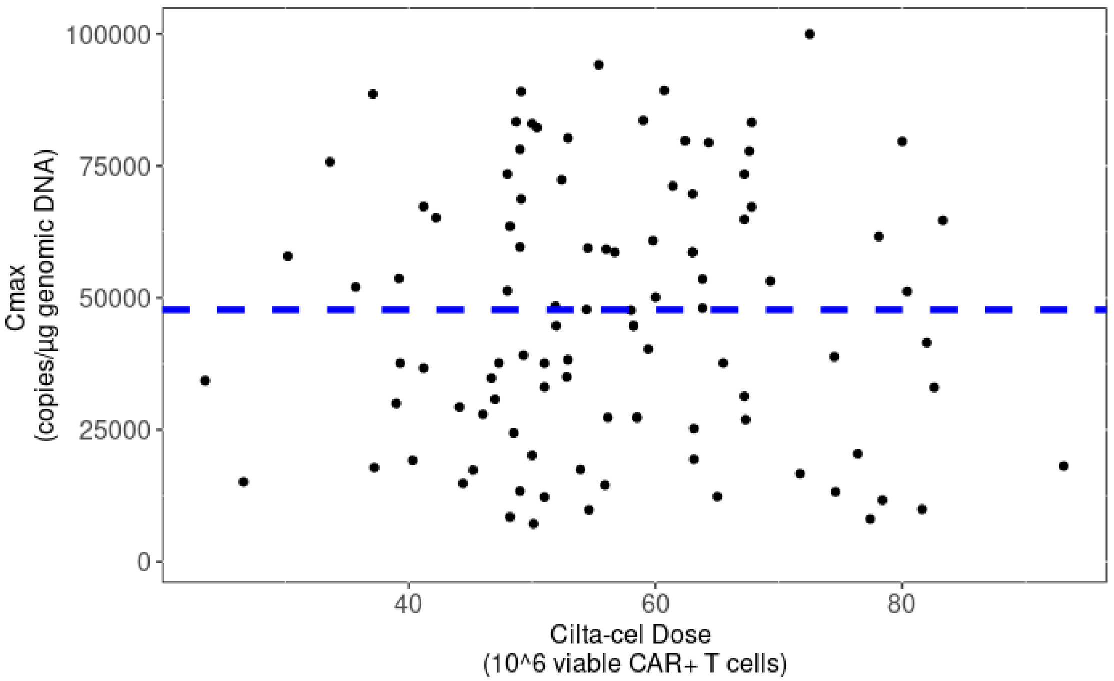
Pharmacokinetics of cilta-cel transgene (mean [SD], t_{max}, t_{las}t and $t_{bql:}$ median [range])	**CART-1** **Phase 1b +2**	**CART-4** **Arm B** **Study treatment**
n	97[a]	176[b]
C_{max}, copies per µg genomic DNA	48692 (27174)	38258 (24133)
t_{max}, day	12.71 (8.73 - 329.77)	12.75 (7.80 - 222.83)
t_{last}, day	125.90 (20.04 - 910.92)	83.01 (12.86 - 630.89)
t_{bql}, day	110.06 (27.89 - 980.87)	100.48 (28.95 - 365.89)
$AUC_{0\text{-}28d}$, day x copies/µg genomic DNA	504496 (385380)	361038 (270918)
$AUC_{0\text{-}6m}$, day x copies/µg genomic DNA	1033528 (1355346)	723717 (736198)
$AUC_{0\text{-}last}$, day x copies/µg genomic DNA	1126574 (1463865)	602674 (862607)
λ_z, day^{-1}	0.0635 (0.0568)	0.0641 (0.0419)
$t_{1/2}$, day	6. (49.8)	21.8 (30.4)

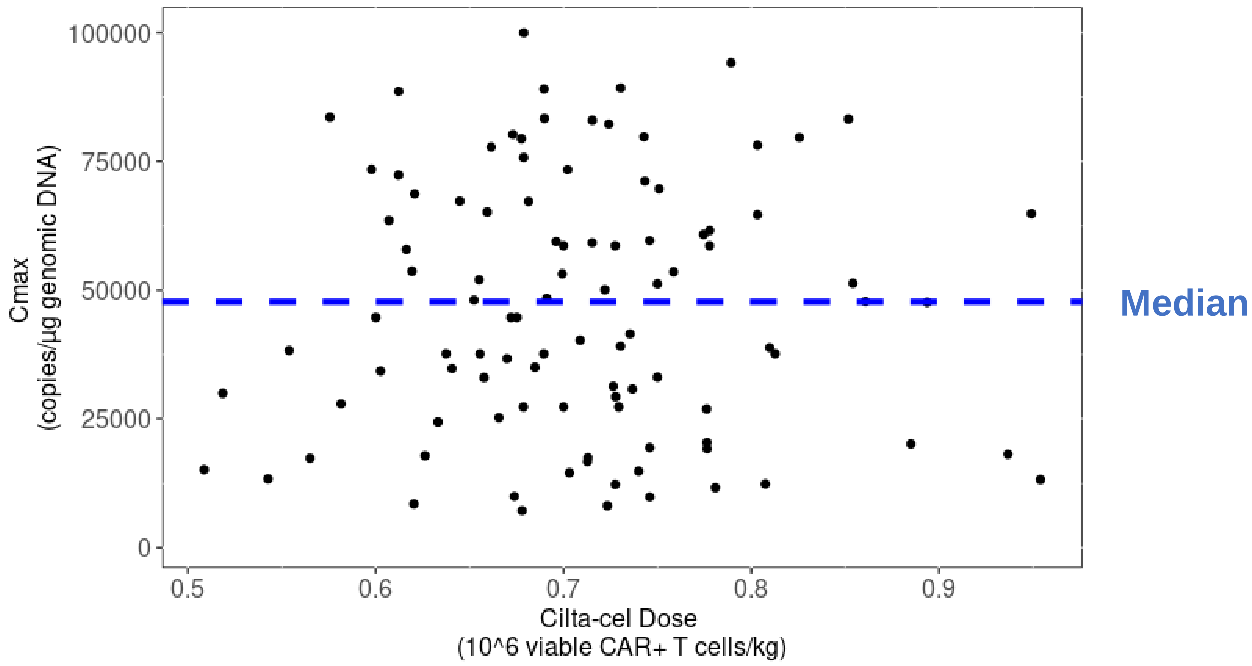
[a]n=96 for $AUC_{0\text{-}6m}$, n=70 for t_{bql}, and n=44 for λ_z, and $t_{1/2}$.
[b]n=175 for C_{max} and t_{max}, n=174 for $_{AUC0\text{-}28d}$, n=115 for $AUC_{0\text{-}6m}$, n=84 for t_{bql}, and n=49 for λz, and t1/2

No Clear Dose-Exposure (C_{max}) Relationship in CART-1

Total targeted dose of 0.75 x 10^6 CAR-positive viable T cells/kg
(range: 0.5-1.0 x 106 CAR-positive viable T cells/kg)

CAR+ T cells Total Dose Administered (Without or With Body Weight Normalization)

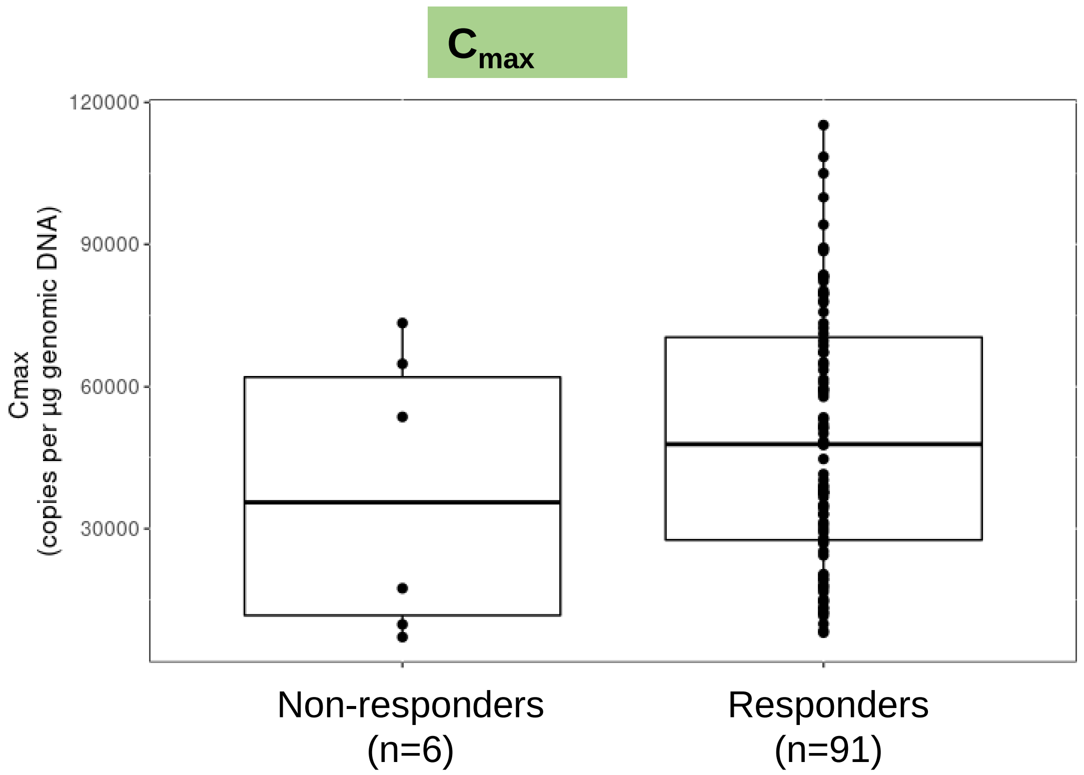


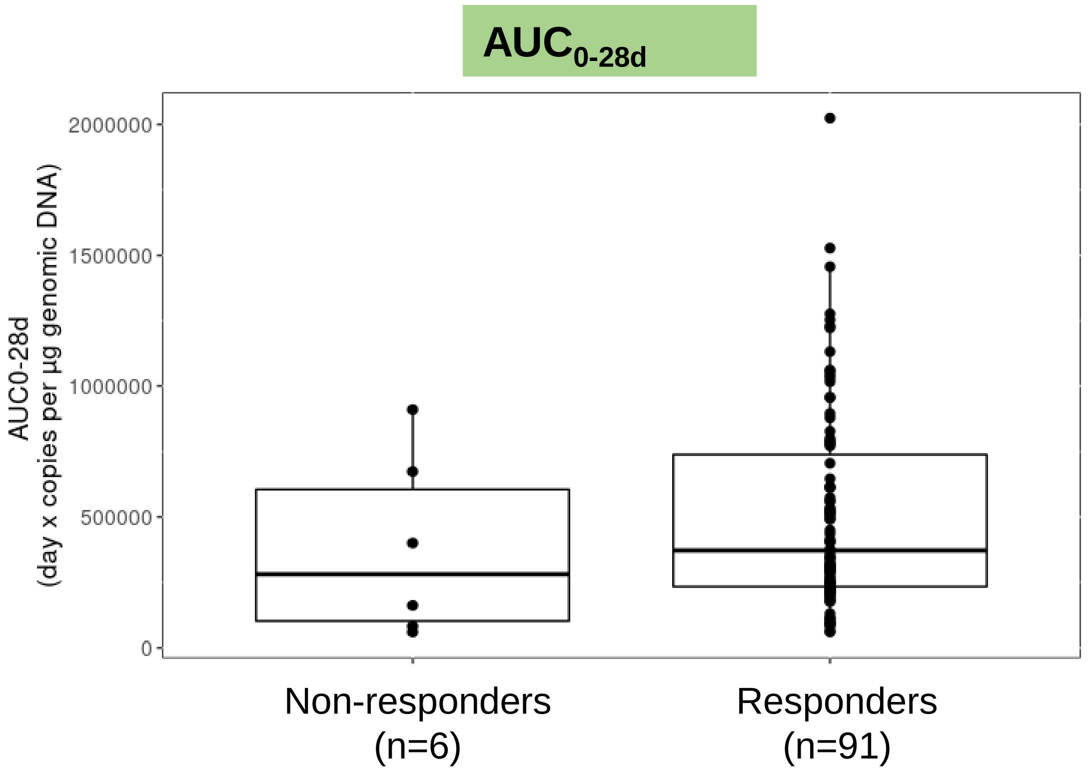


Chang., et al., ACCP poster 2023

ER for Efficacy in CART-1

Scatter-Boxplot of Blood PK transgene (C_{max} and $AUC_{0\text{-}28d}$) by Non-responders vs. Responders

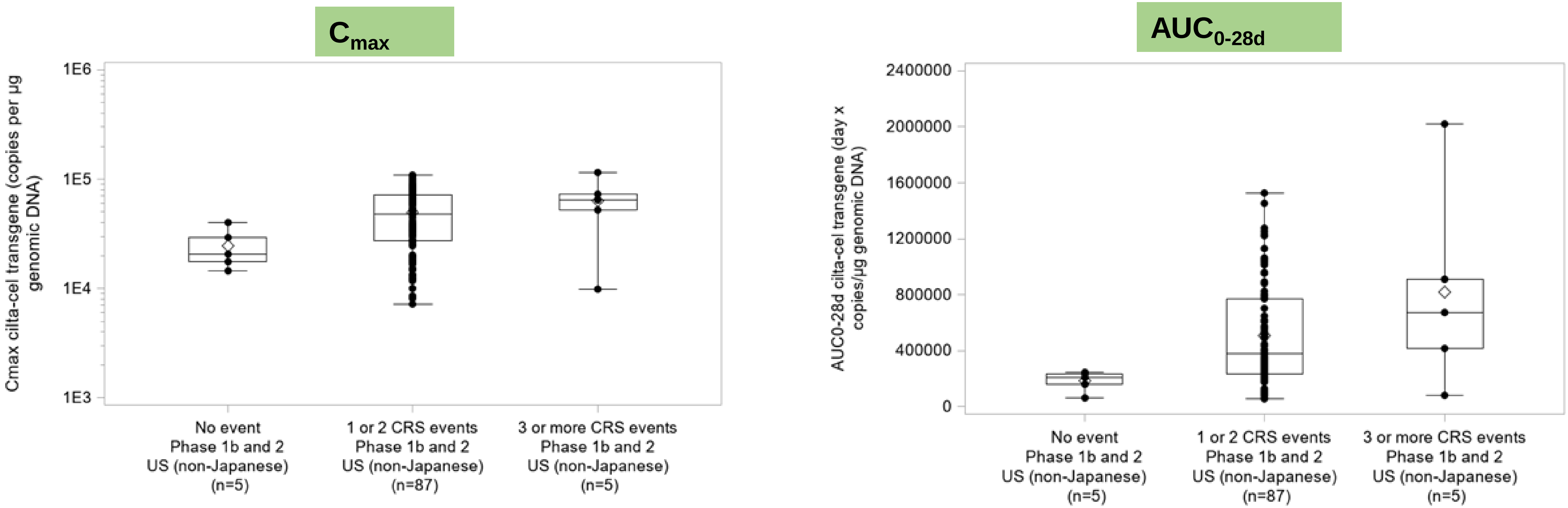




- Higher median PK parameters in responders (sCR+CR+VGPR+PR) compared to non-responders but the ranges were overlapping
- Limited sample size in non-responders (n=6); data was skewed

ER for Safety in CART-1

CRS: cytokine release syndrome

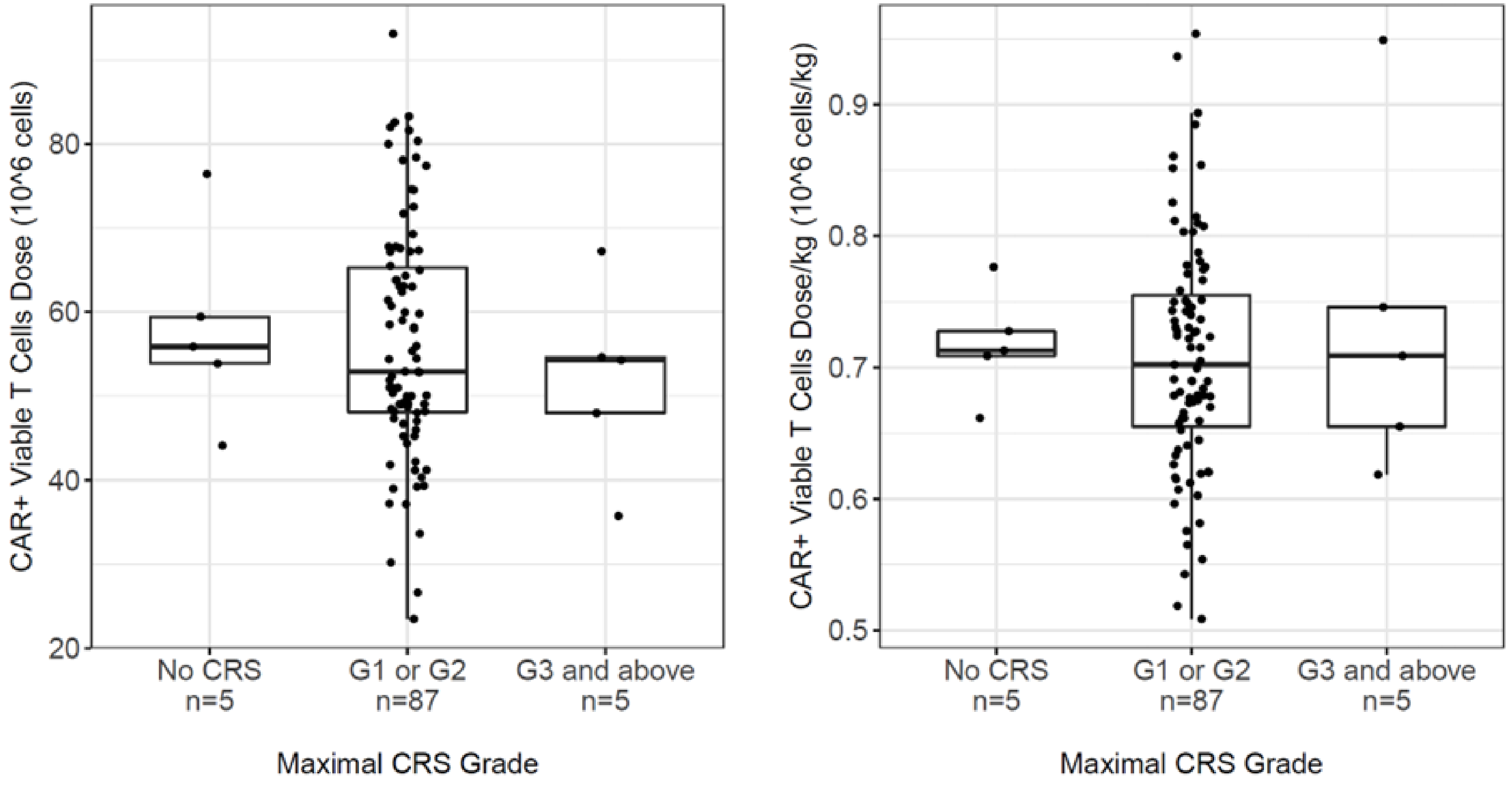


- All-grade CRS was observed for 92 subjects (94.8%)
- Apparent trend of higher exposure during the expansion phase: higher medium PK during expansion but the ranges were overlapping
- Higher median C_{max} and AUC_{0-28d} among subjects who were treated with <u>tocilizumab/corticosteroids</u>-confounding concurrence of CRS and overlapping exposure range.

ER for Safety in CART-1

CRS: cytokine release syndrome

CAR+ T cells Total Dose Administered (With or Without Body Weight Normalization)



The lack of relationship with dose is expected since only one target dose level (0.75 [range: 0.5-1.0] x 10^6 CAR-positive viable T cells/kg) of cilta-cel was investigated

ER for Safety in CAR-T1

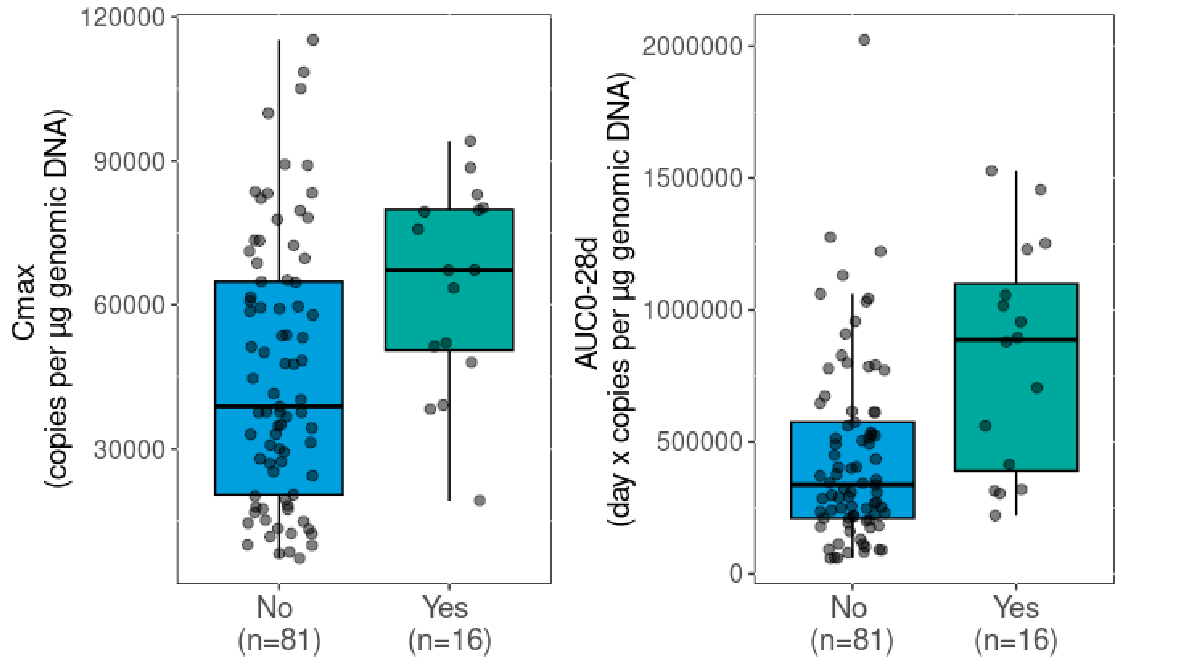
Nes: neurologic events

*: Immune effector cell-associated neurotoxicity syndrome:
**: Other neurotoxicity

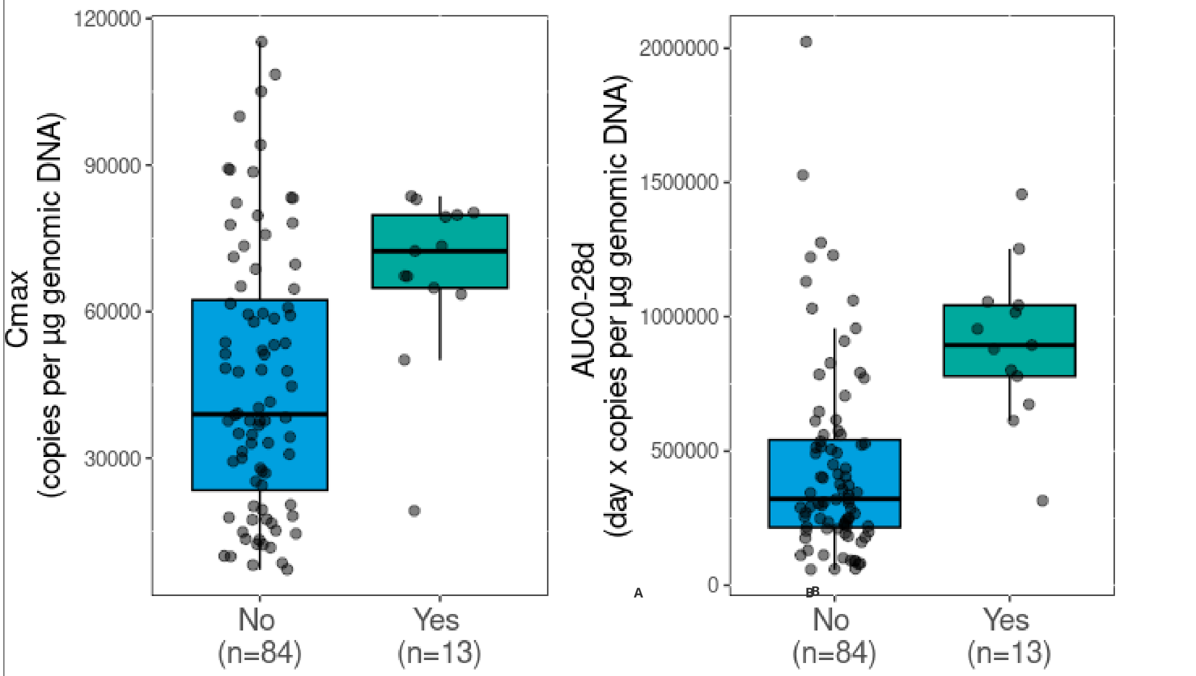
ICANS (n=16, including 2 subjects with Grade 3 or 4)

Other neurotoxicity: ONTX (n=13, including movement and neurocognitive treatment-emergent adverse events: MNTs (n=6): 3 subjects with Grade 2, 9 subjects with Grade 3 or 4, and 1 subject with Grade 5)

Scatter-Boxplot of Blood PK transgene (C_{max} and $AUC_{0\text{-}28d}$) by ICANS



Scatter-Boxplot of Blood PK transgene (C_{max} and $AUC_{0\text{-}28d}$) by ONTX



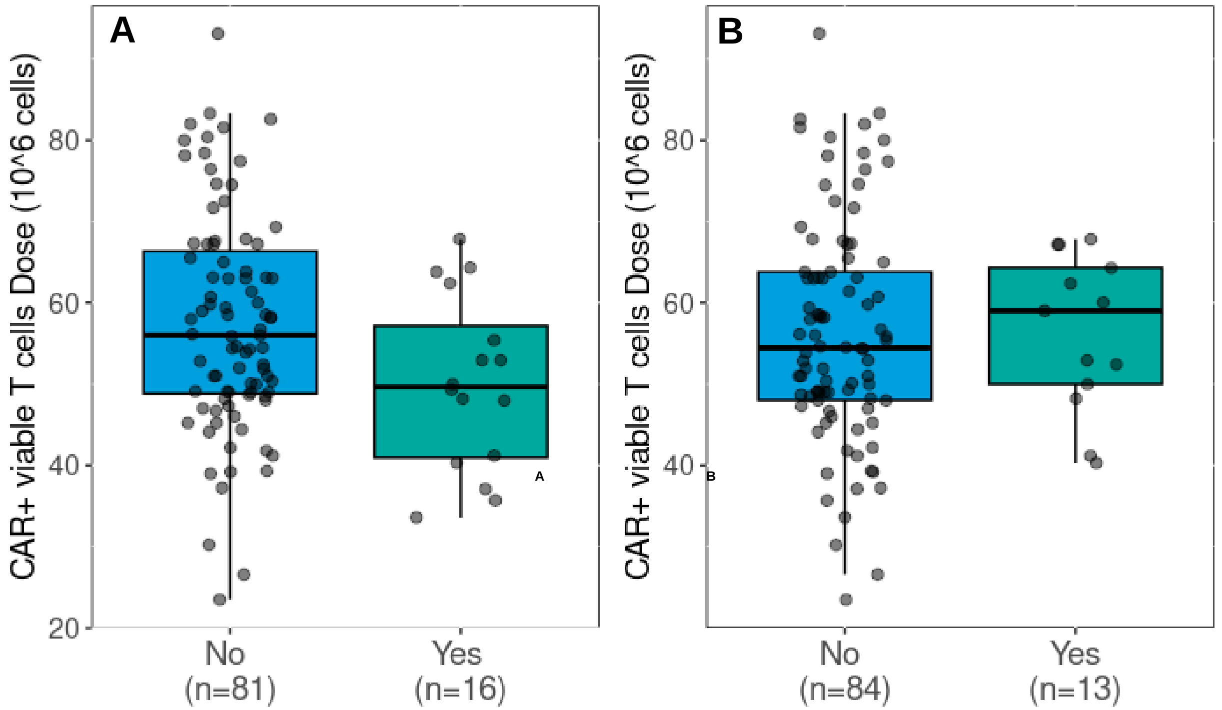
- Subjects with CAR-T cell neurotoxicity had apparent higher median CAR transgene PK metrics (C_{max} and $AUC_{0\text{-}28d}$) compared to subjects without neurotoxicity AEs
- The association between exposure and neurotoxicity AEs was not clear due to wide overlapping ranges of PK transgene levels and high interindividual variability

Chang., et al., ACCP poster 2023

ER for Safety in CAR-T1

NEs: neurologic events

Scatter-Boxplot of Dose Administrated by AE status (A: ICANS and B: ONTX)

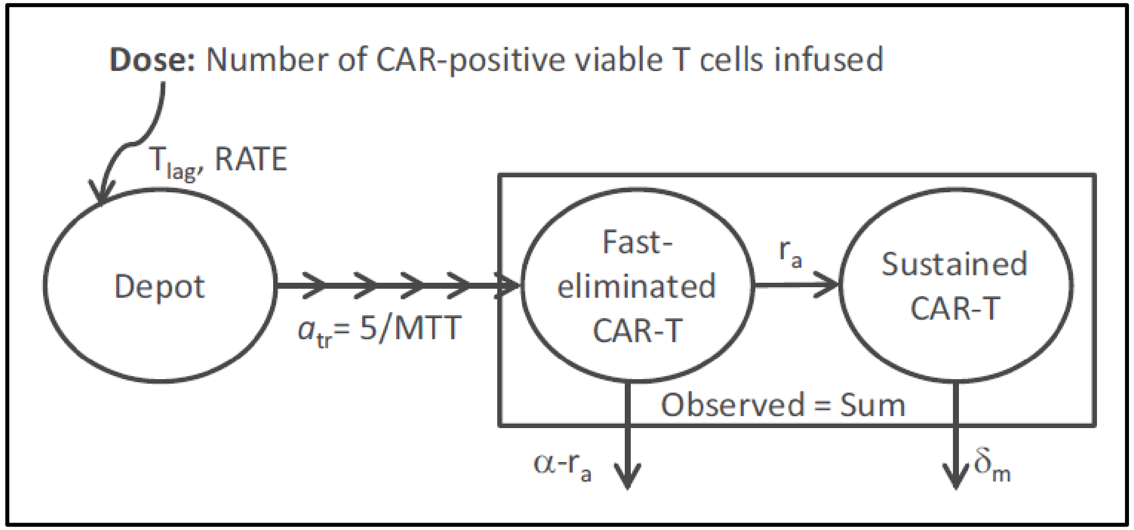


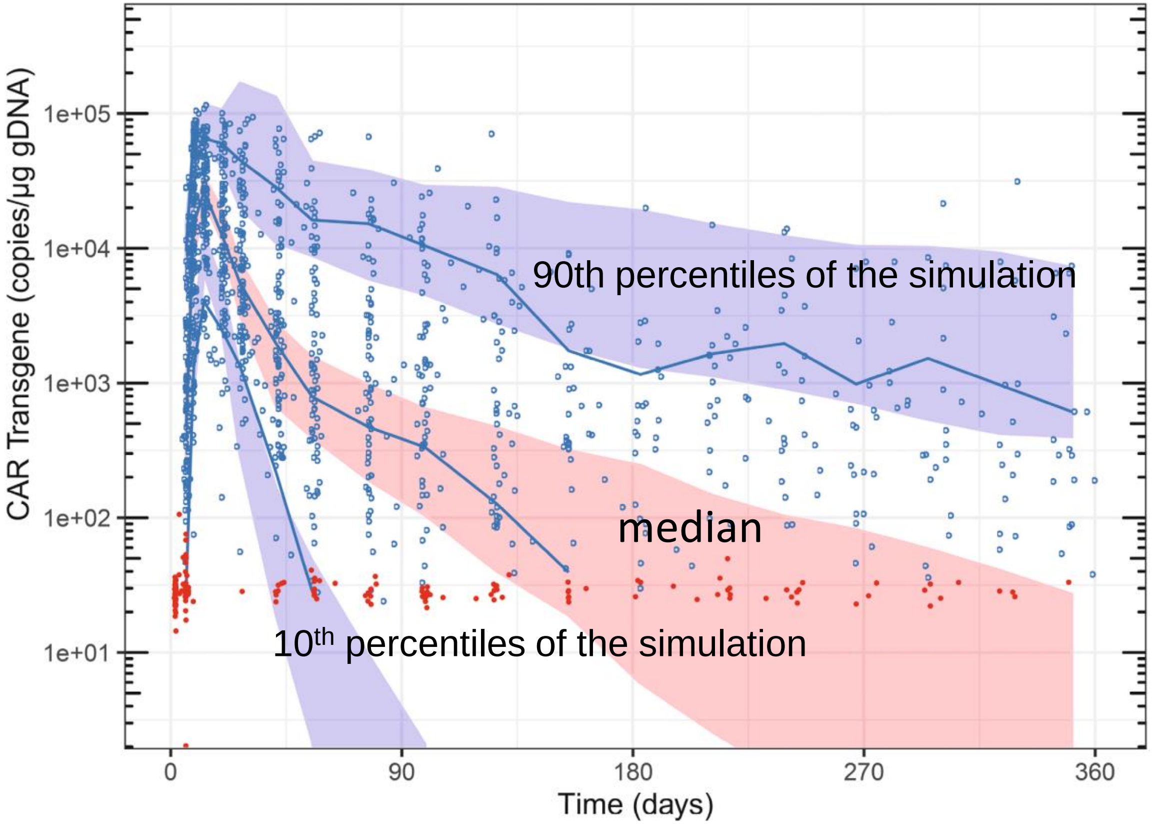
There was no association between total cilta-cel dose administrated and ICANS or ONTX

Chang., et al., ACCP poster 2023

PopPK Model Based on Transgene Results

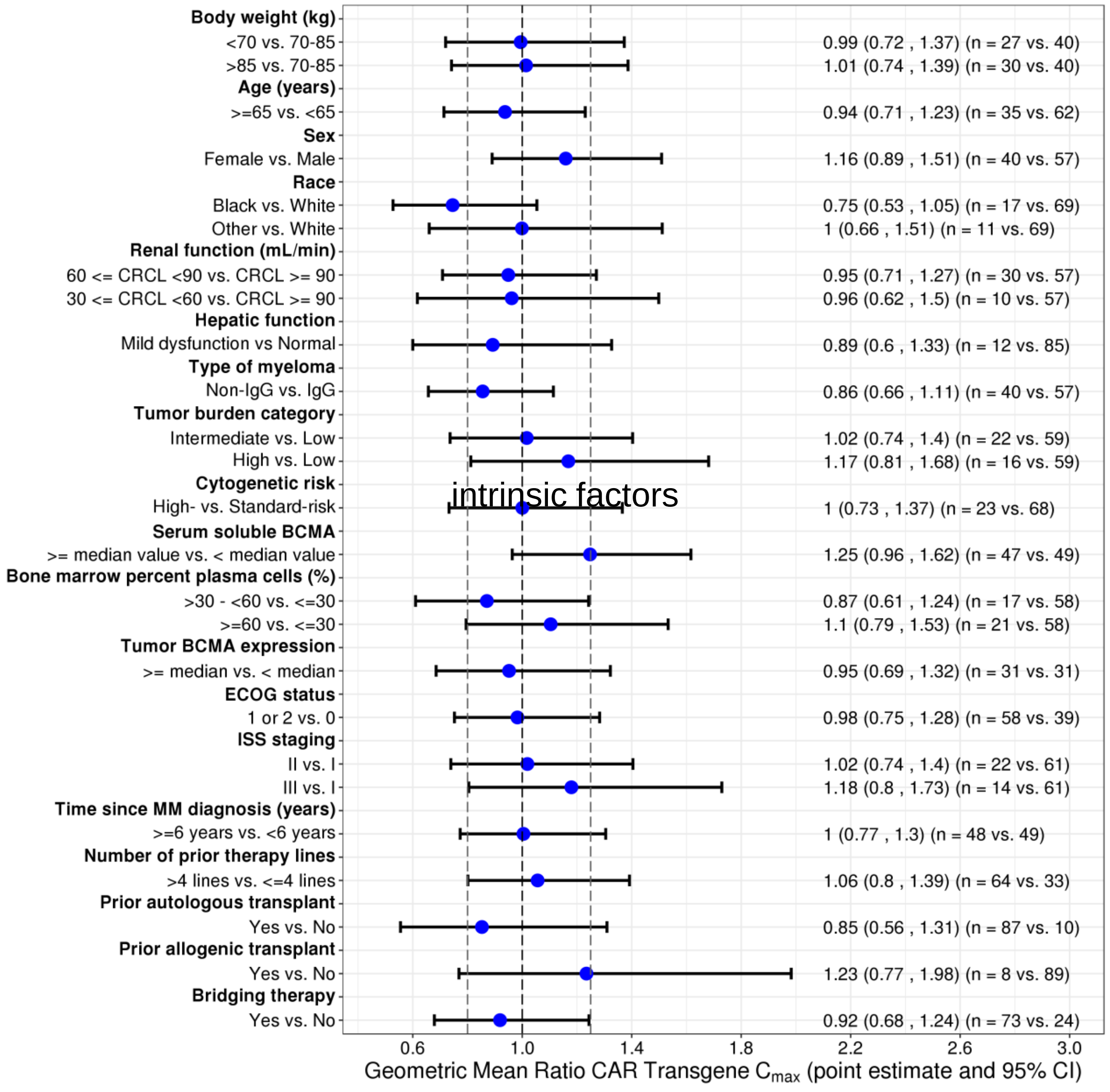
Empirical 2-compartment model (not mass-balance):
Dose: number of CAR-positive viable T cells infused
Fitted in: Observed CAR transgene level (copies/µg gDNA)





Parameters: lag time (Tlag)
mean transit time (MTT),
apparent rapid decline rate (α, apparent transition rate from fast eliminated to sustained CAR-T (ra)
apparent gradual decline rate ((δ_m).
initial margination (distribution), expansion, differentiation, contraction, and persistence.

Covariates of Interests on Predicted Transgene C_{max}



Intrinsic factors

None of the estimated geometric mean ratio confidence intervals (CI) excludes the null value (ie, geometric mean ratio of 1)

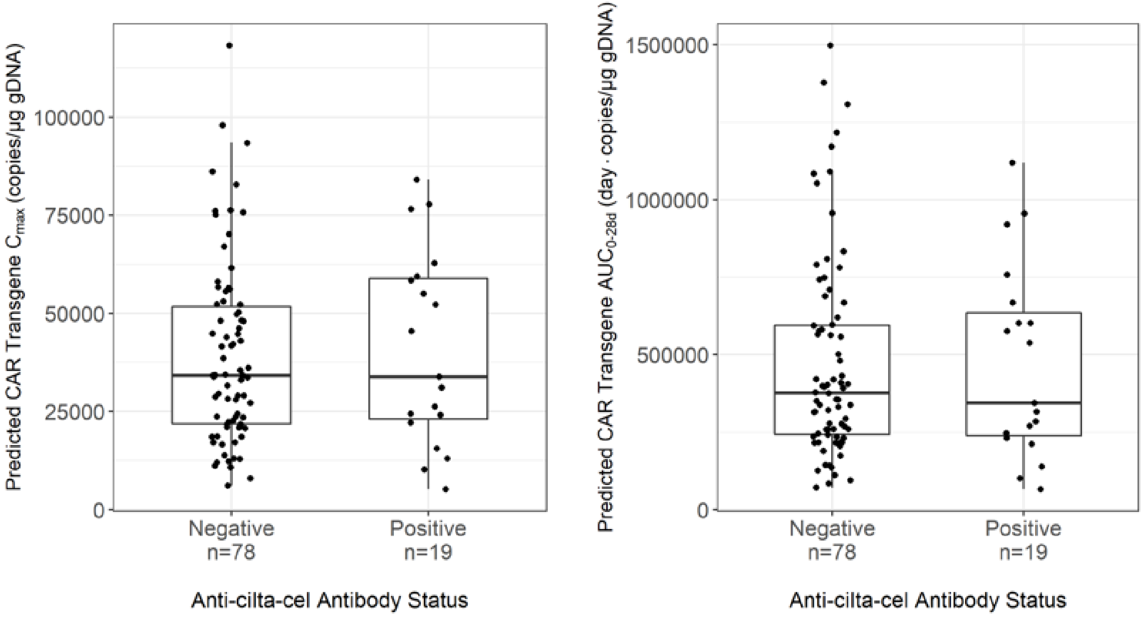
Extrinsic factors such as manufacturing process change can be tested too

Q: Does immunogenicity (ADA-mediated immune response) impact on cilta-cel exposure (cell expansion)?

Immunogenicity (ADA) and PK

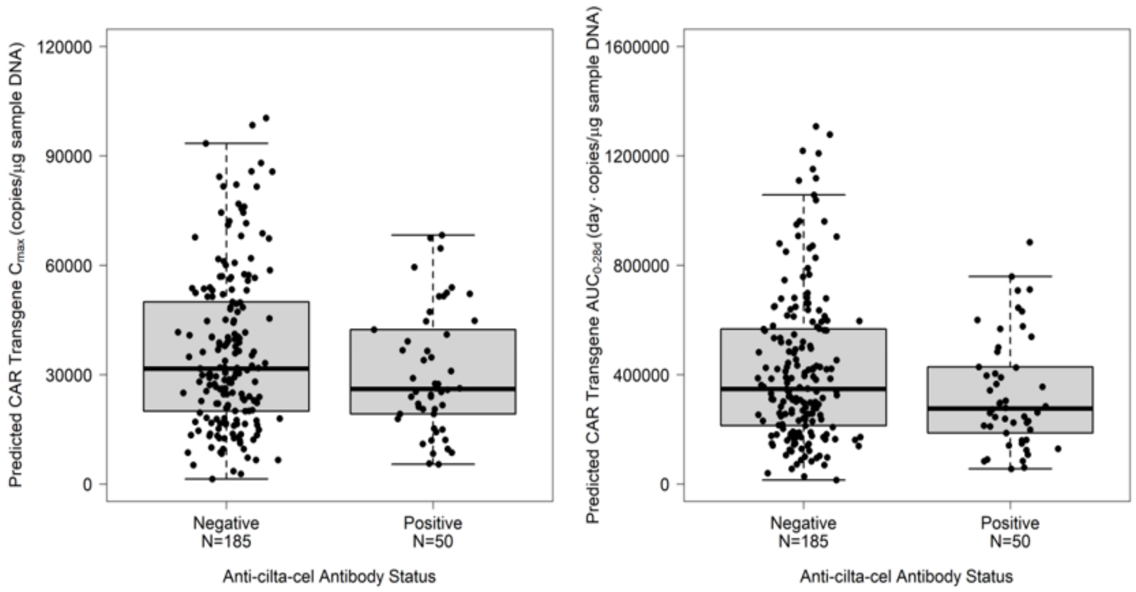
CART-1 (MMY2001)

Scatter-Boxplot of Blood PK transgene (C_{max} and AUC_{0-28d}) by ADA



CART-4 (MMY3002)

Scatter-Boxplot of Blood PK transgene (C_{max} and AUC_{0-28d}) by ADA

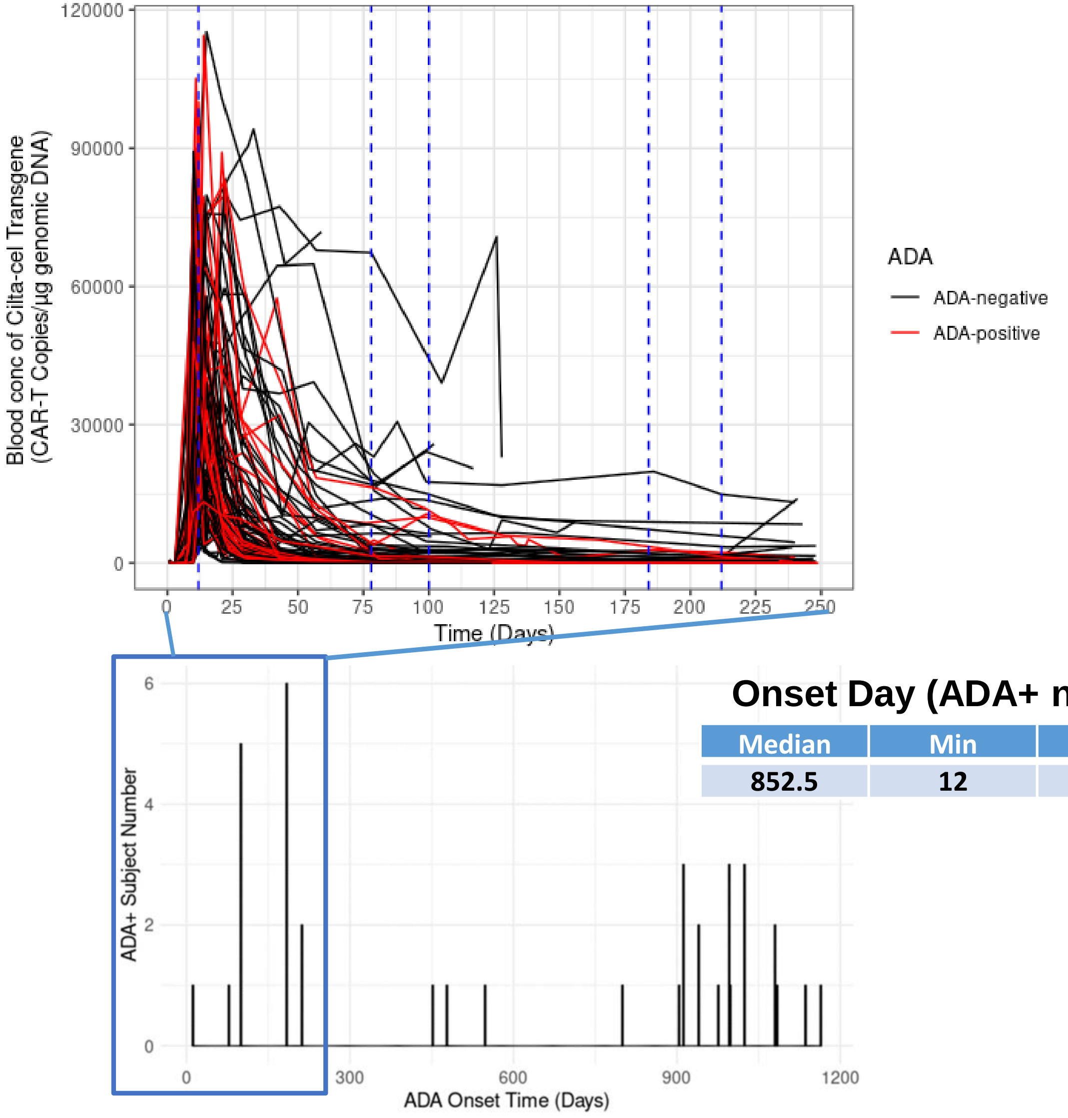


No apparent difference of CAR transgene exposure (i.e., predicted C_{max} and AUC_{0-28d}) was observed between ADA+ and ADA- subjects in CART-1 and CART-4

ADA Incidence and Onset Time in CART-1 and CART-4

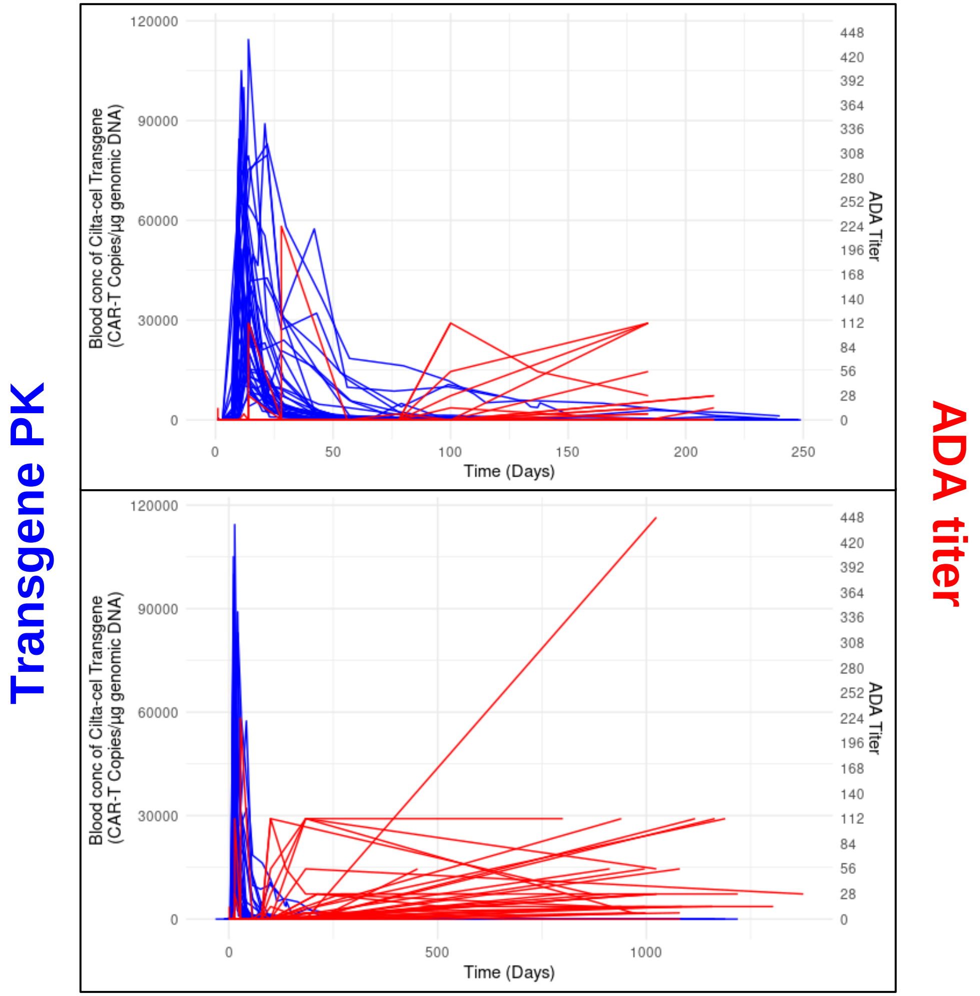
Study Name	Issued CSR (CCO)	ADA incidence	Median time (Days) to first ADA+ (Min- Max)
CARTITUDE-1 (68284528MMY2001) US	Primary analysis CSR SCP addendum Closeout CSR	15.5% 19.6% (used in label) 39.2%	114.01 (10.64-197.79) 852 (12 -1164)
CARTITUDE-4 (68284528MMY3002)	Primary analysis CSR	21.0%	111.89 (56.75-362.96)

Majority of ADA+ Had Late Onset Time (Not in Expansion Phase) and Low Titers in CART-1 Closeout



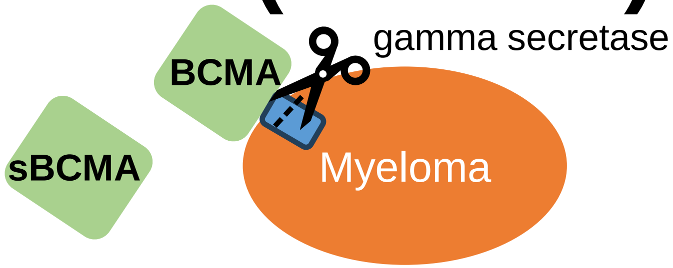
Onset Day (ADA+ n= 37)

Median	Min	Max
852.5	12	1164



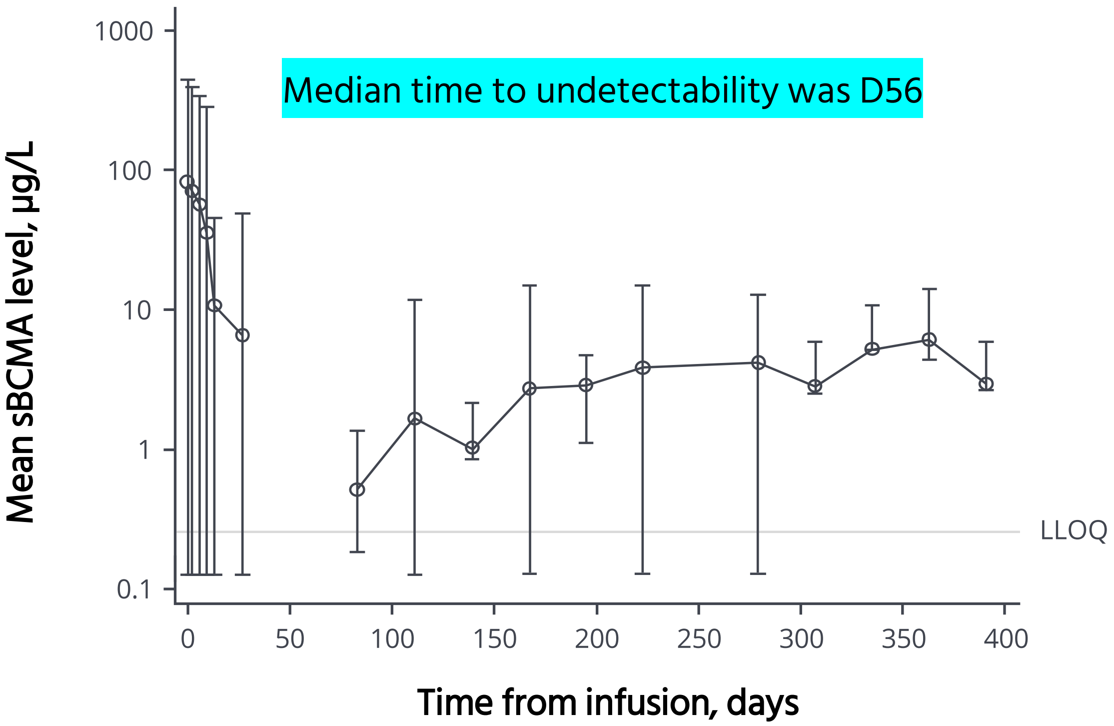
There is no evidence to show that immunogenicity has an apparent impact on cell expansion (exposure), safety (related AE), and efficacy.

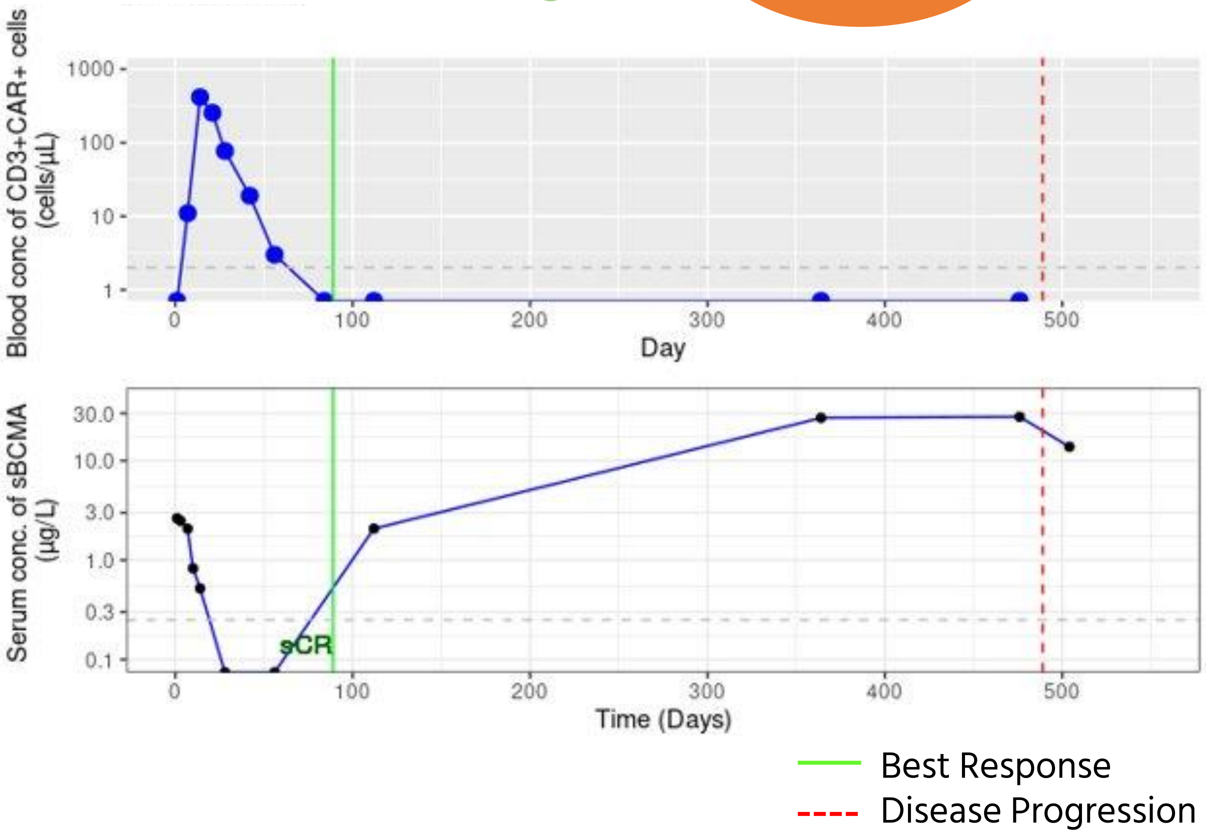
Pharmacodynamic Marker: Soluble BCMA (sBCMA)



- The value of sBCMA as biomarker for disease progression in the clinic

Mean sBCMA levels over time in CART-4 (MMY3002)

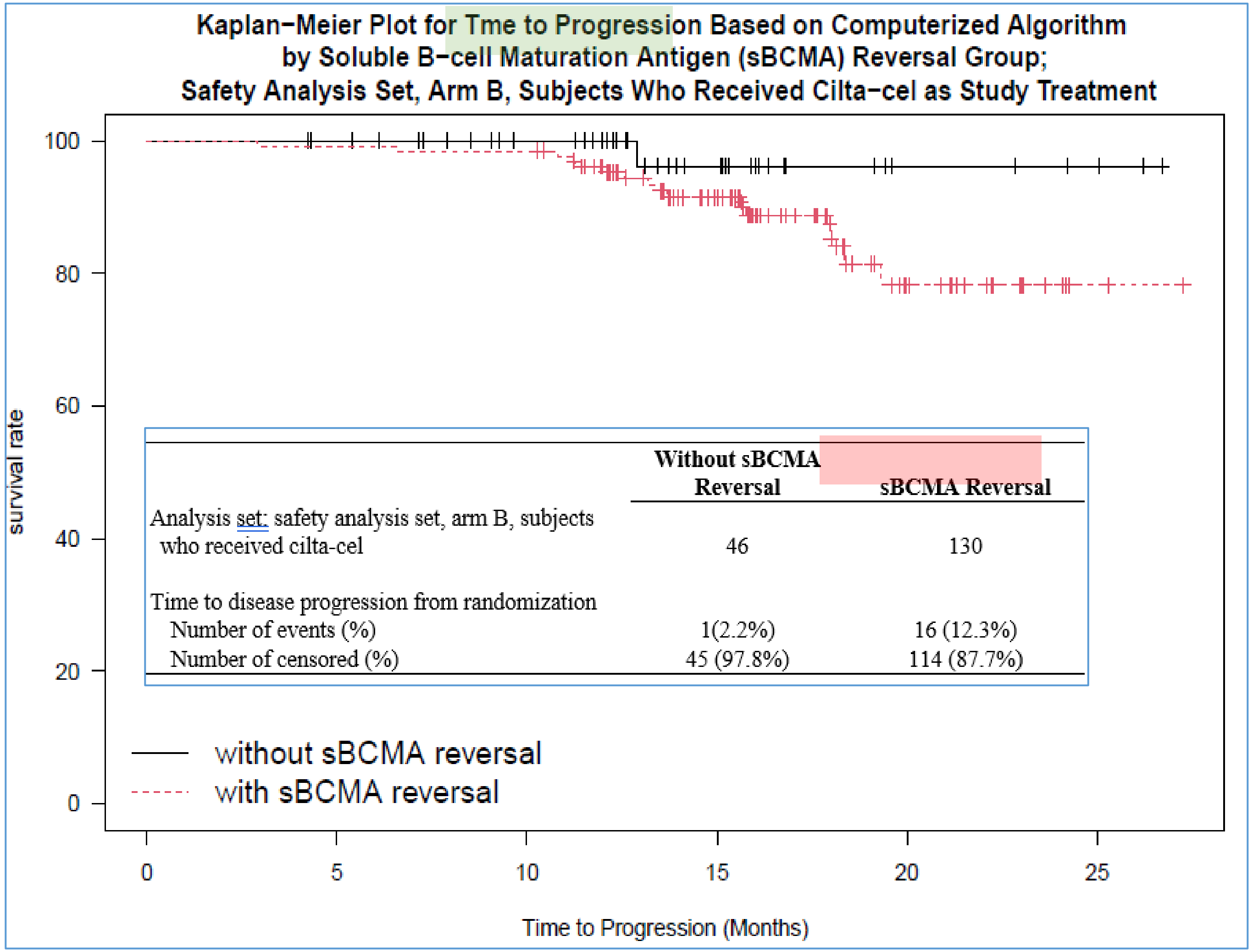




Mean conc was below LLOQ (bql = 0) on Day 56, suggesting sBCMA reached to nadir
After Day 56, some subjects had higher sBCMA conc during disease progression (sBCMA reversal vs. disease progression)

CF de Larrea., et al., IMS 2023

Pharmacodynamic Marker: sBCMA vs. Time to Event Methods (e.g., Kaplan–Meier)

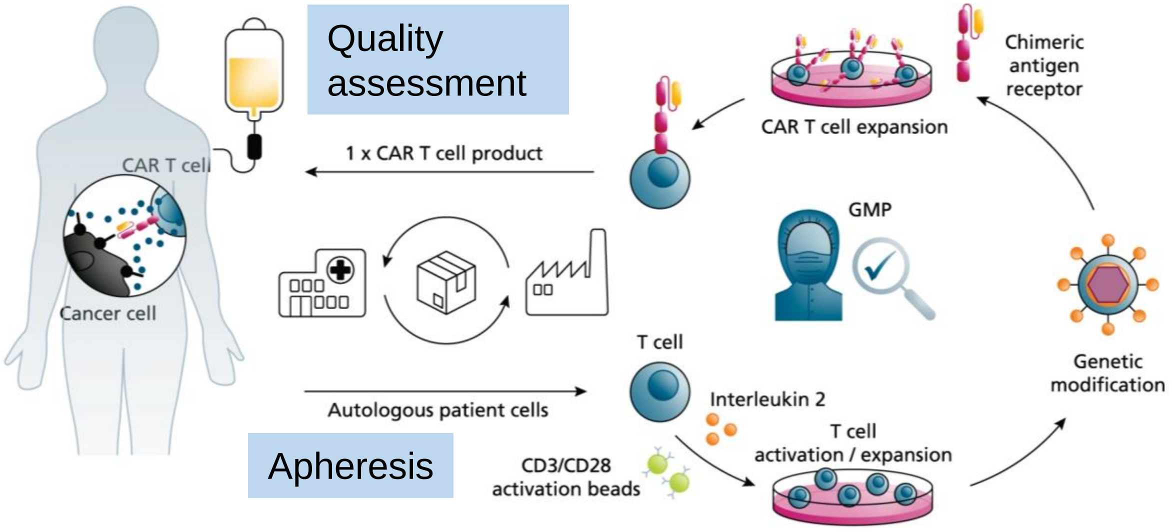


This retrospective analysis (subgroup) indicates that PD modeling is required to further validate the hypothesis that sBCMA can serve as a clinical biomarker for disease progression.

Challenges and Opportunities in CAR T Products (Personalized Medicine)

- CMC issue (QC issue, limited manufacturing capacity, lentivirus manufacturing process, OOS comparability studies)

- The mechanism of NE is unclear (biodistribution)
  - Radiolabel (half-life is too short) or reporter gene transfected- CAR+ cells to figure out bio-distribution



Blache, U. et al., Nat Commun 13, 5225 (2022).

Q: How to optimize the manufacturing process and accelerate the development of cell therapy?

A: Translational research (T-cell population and phenotyping over time), cross-functional collaboration, and industry and health authority collaboration

Case Study: CARVYKTI®
(ciltacabtagene Autoleucel;Cilta-cel)

CARTITUDE-1 (MMY2001: phase1b/2) and CARTITUDE-4 (MMY3002: Phase3) for BLA

- The PK profiles of cilta-cel, characterized by significant interindividual variability, displayed distinct phases of expansion, contraction, and persistence.
- Exposure-response analysis (efficacy/safety): a trend of higher C_{max}/$AUC_{0\text{-}28d}$ was observed but wide overlapping ranges. The dose-exposure relationship was not clear
- Immunogenicity (ADA) doesn’t have an apparent impact on exposure, safety, and efficacy (provides a rationale for not performing neutralizing antibody assessments)
- sBCMA appears to be a promising pharmacodynamic (PD) biomarker, particularly when combined with M-protein, for potential clinical applications in the future.
    - Further exploratory PD modeling is warranted to investigate its full potential.

Acknowledgements:

JnJ clinical colleagues, especially Translational Research (Biomarker)

Legend colleagues

Health providers/Investigators

Our patients and their families

Thank You
Any Questions?